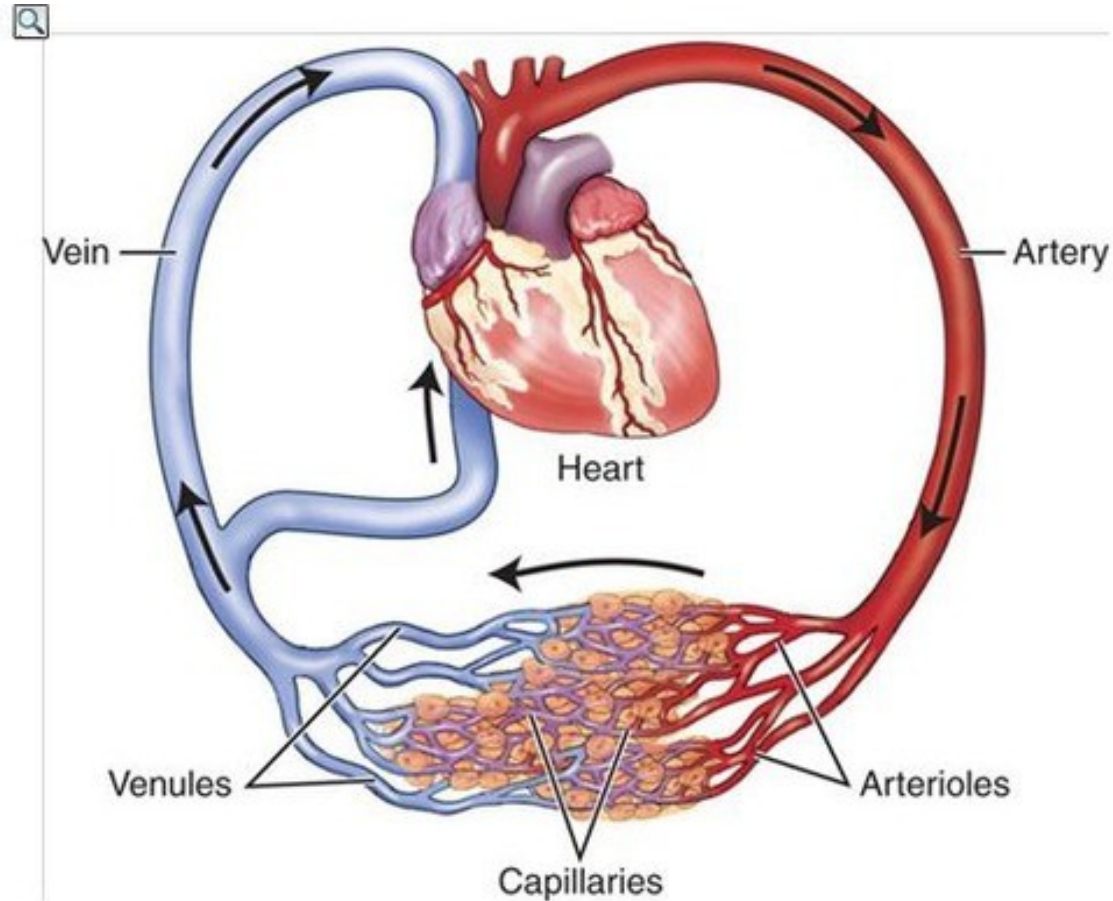


Blood vessels of the body



By Diliab D.

Outline

- **Principal Arteries of the body**
- **Venous drainage of the body**
- **Lymphatic drainage of the body**

Principal Arteries of the body

The Aorta

- The aorta is the largest artery of the body, with a diameter of 2–3 cm.
- Its four principal divisions are the **ascending aorta, arch of the aorta, thoracic aorta, and abdominal aorta.**
- The portion of the aorta that emerges from the left ventricle posterior to the pulmonary trunk is the ascending aorta.
- The ascending aorta gives off two **coronary artery branches** that supply the heart.
- Then the ascending aorta arches to the left, forming the arch of the aorta, which descends and ends at the level of the IVD between the 4th and 5th thoracicvertebrae.

Principal Arteries of the body

The Aorta...

- The section of the aorta between the arch of the aorta and the diaphragm is called the **thoracic aorta**
- The section between the diaphragm and the common iliac arteries is the **abdominal aorta**.
- As the aorta continues to descend, it lies close to the vertebral bodies, passes through the **aortic hiatus** of the diaphragm, and divides at the level of the **4th lumbar** vertebra into **two common iliac arteries**, which carry blood to the lower limbs.
- Each division of the aorta gives off arteries that branch into distributing arteries that lead to various organs.
- Within the organs, the arteries divide into arterioles and then into capillaries that service the systemic tissues.

Principal Arteries of the body

Aorta

❖ Ascending aorta

- ✓ ascends from the heart (left ventricle)
- ✓ The **coronary arteries** are the **only branch** of the ascending aorta which supplies the heart

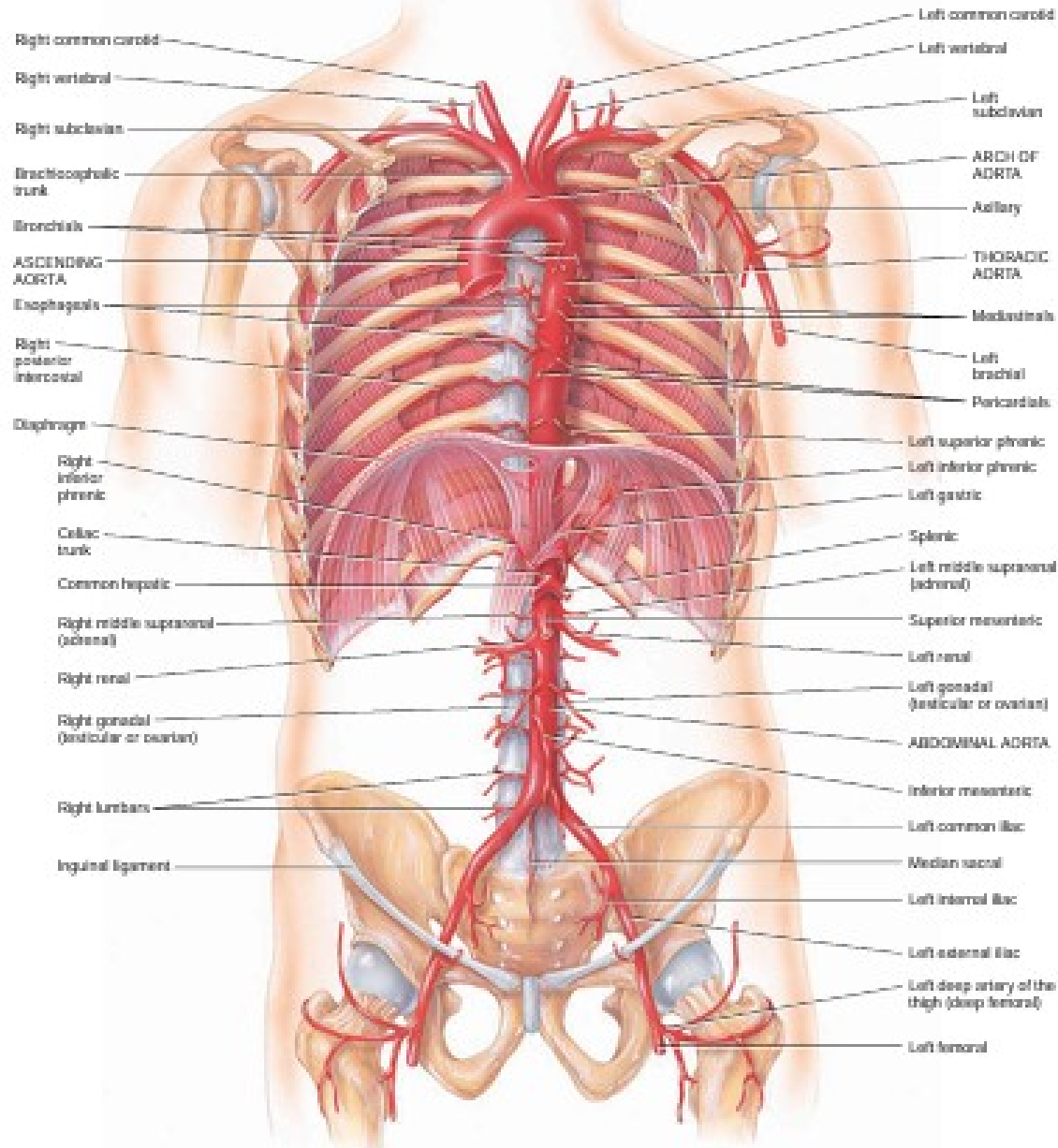
❖ Aortic arch

- ✓ The curved continuation of the ascending aorta
- ✓ Three vessels arise from the aortic arch:
 - **Brachiocephalic artery**– further bifurcates into **Right subclavian** and **common carotid** a that supply the right upper limbs and head regions
 - **Left common carotid artery**
 - **Left subclavian artery**
- ✓ The arch ends by becoming the thoracic(descending) aorta posterior to the **2nd left sternocostal joint**.

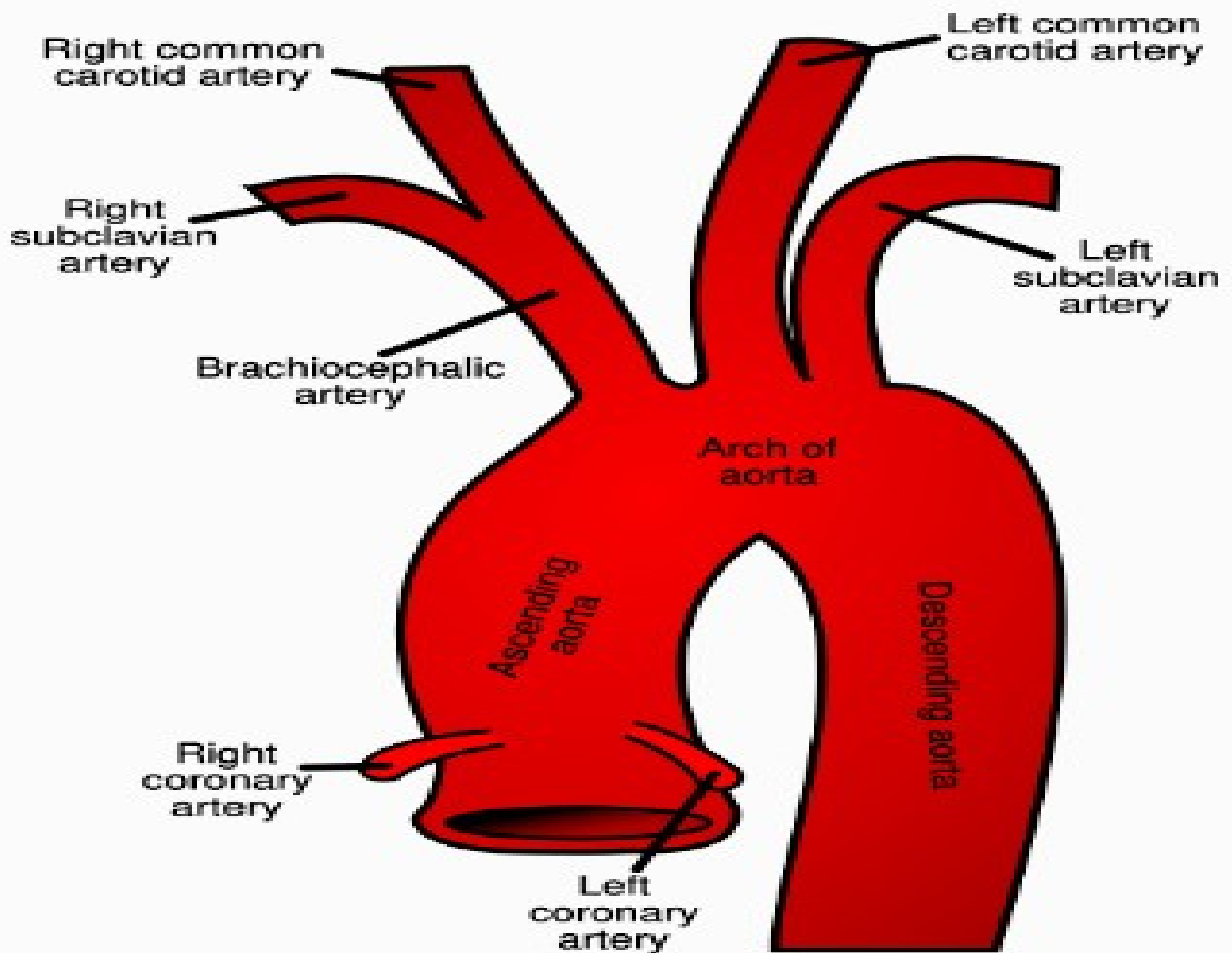
❖ Descending aorta

- ✓ Thoracic aorta (in the chest) and
- ✓ Abdominal aorta (in the abdomen)

Principal Arteries of the body



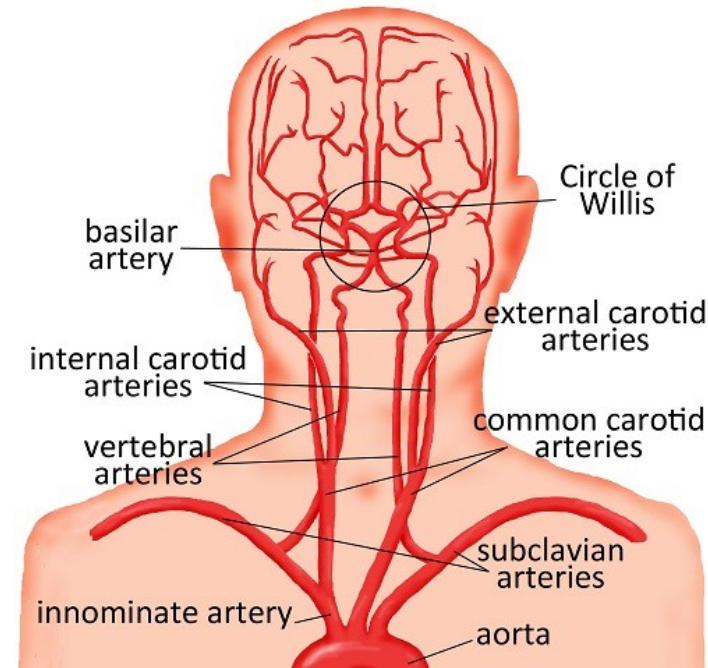
(b) Detailed anterior view of the principal branches of the aorta



Arteries of the neck and head

■ *Common carotid artery*

- Has asymmetrical origin
 - Brachiocephalic trunk (R)
 - Arch of aorta (L)
- Ascends upwards in the neck lateral to trachea
- Divides slightly below the angle of the mandible into
 - **Internal carotid artery** and
 - **External carotid artery**

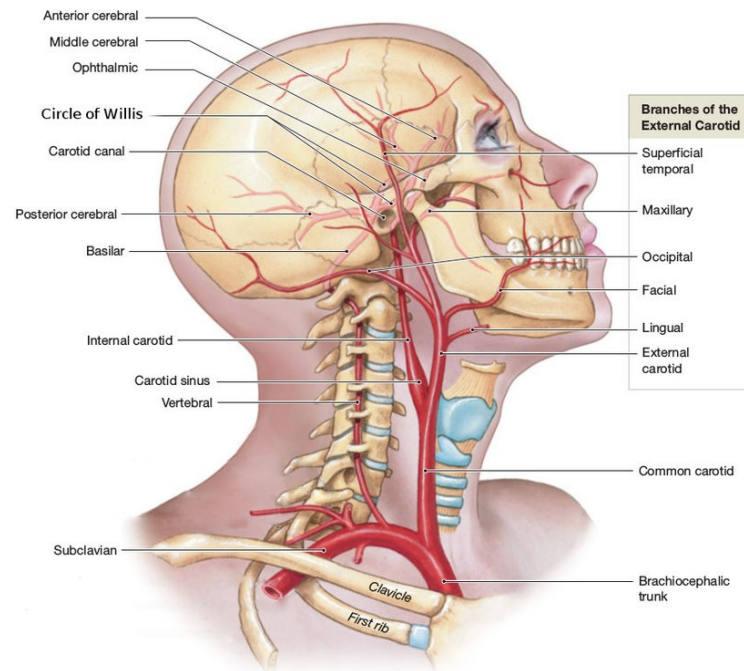


Common carotid artery...

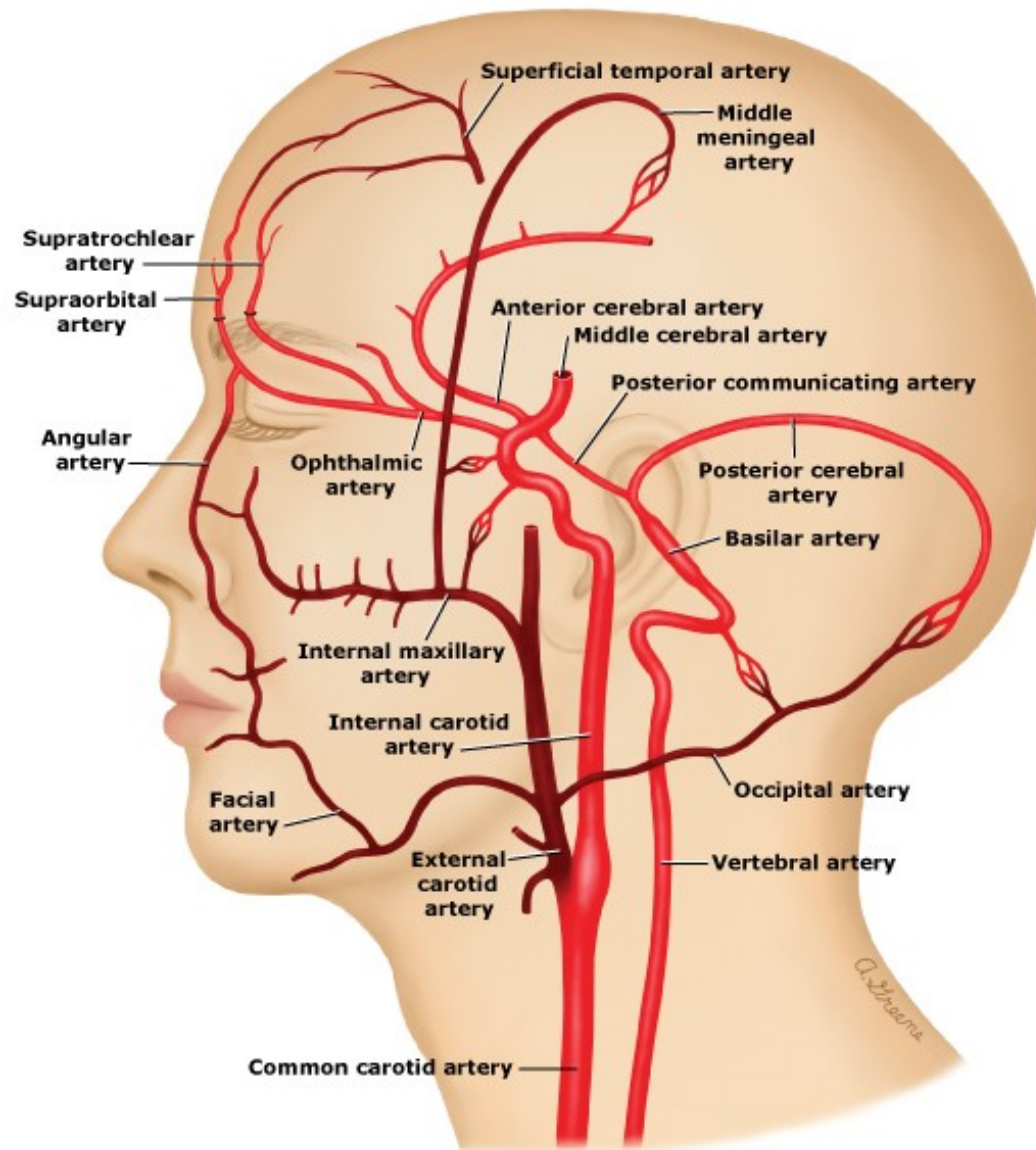
○ The internal carotid arteries

- ✓ are direct continuations of the common carotids
- ✓ It enter the cranium through the **carotid canals** in the temporal bones and become the main arteries of the **brain** and structures in the **orbits**
- ✓ No named branches arise from the internal carotid arteries in the neck.
- ✓ Terminal branches of internal carotid arteries are

- **Anterior cerebral artery**- supplies most of the **medial surface of the cerebrum** and deep masses of **gray matter** within the cerebrum,
- **Middle cerebral artery**- which supplies most of the lateral surface of the cerebrum

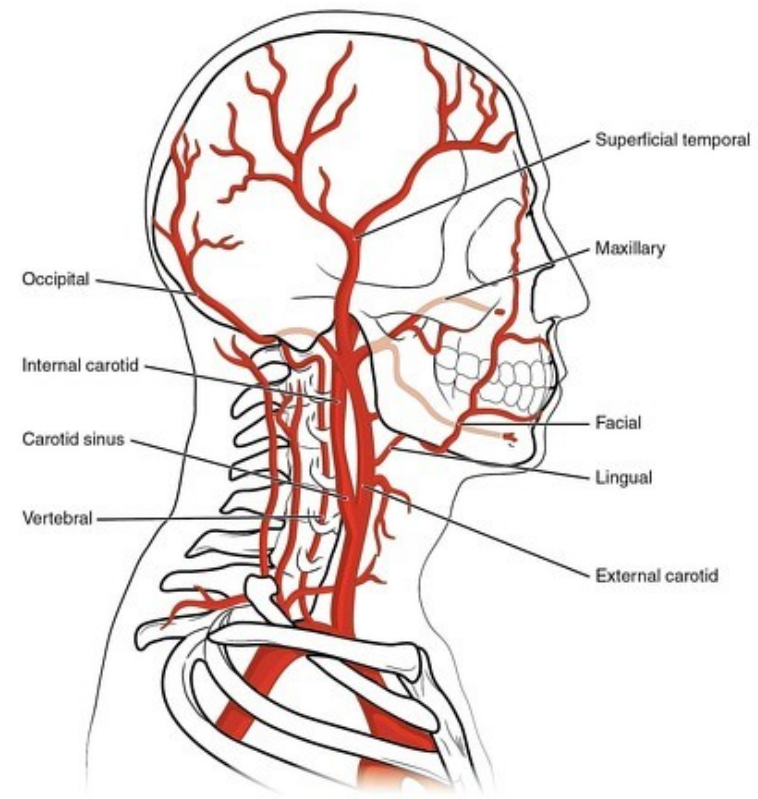


Internal carotid arteries...



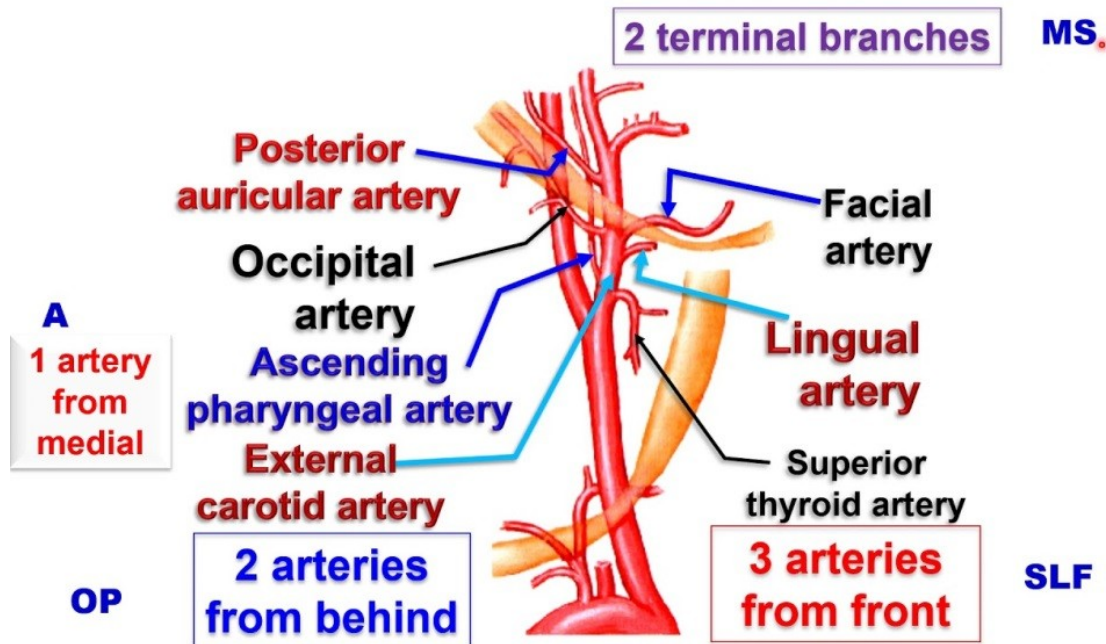
○ The external carotid

- ✓ Supply most structures **external the cranium**
- ✓ two terminal branches,
 - **Maxillary artery-** supply **teeth**, gums, muscles of mastication, nasal cavities, eyelids
 - **Superficial temporal-** supply parotid gland, side of the head
- ✓ Before these terminal branches, **six arteries** arise from the external carotid artery



Branches external carotid

1. **Ascending pharyngeal artery:**
 - supply pharyngeal area
2. **Occipital artery:**
 - supply scalp (posteriorly), meninges, some of the posterior neck muscles



Branches external carotid...

3. Posterior auricular artery:

- Supply parotid gland, auricle, & scalp

4. Superior thyroid artery:

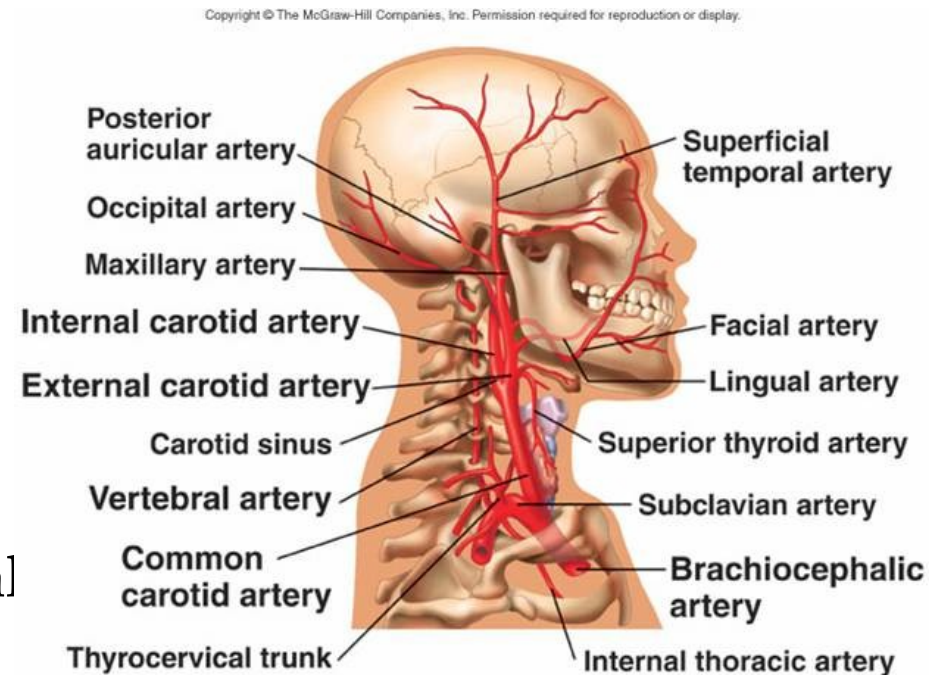
- Supply thyroid gland, infrahyoid muscles and SCM and larynx.

5. Lingual artery:

- Supply tongue and sublingual gland

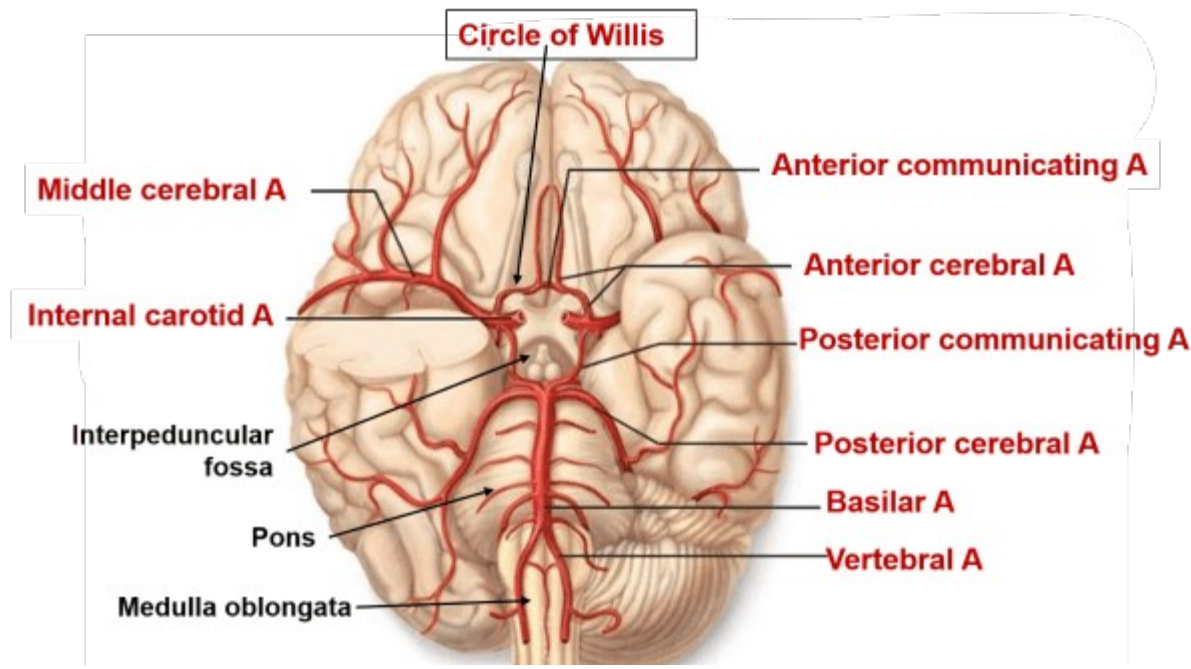
6. Facial artery:

- Supply palate, chin, lips, nasal region

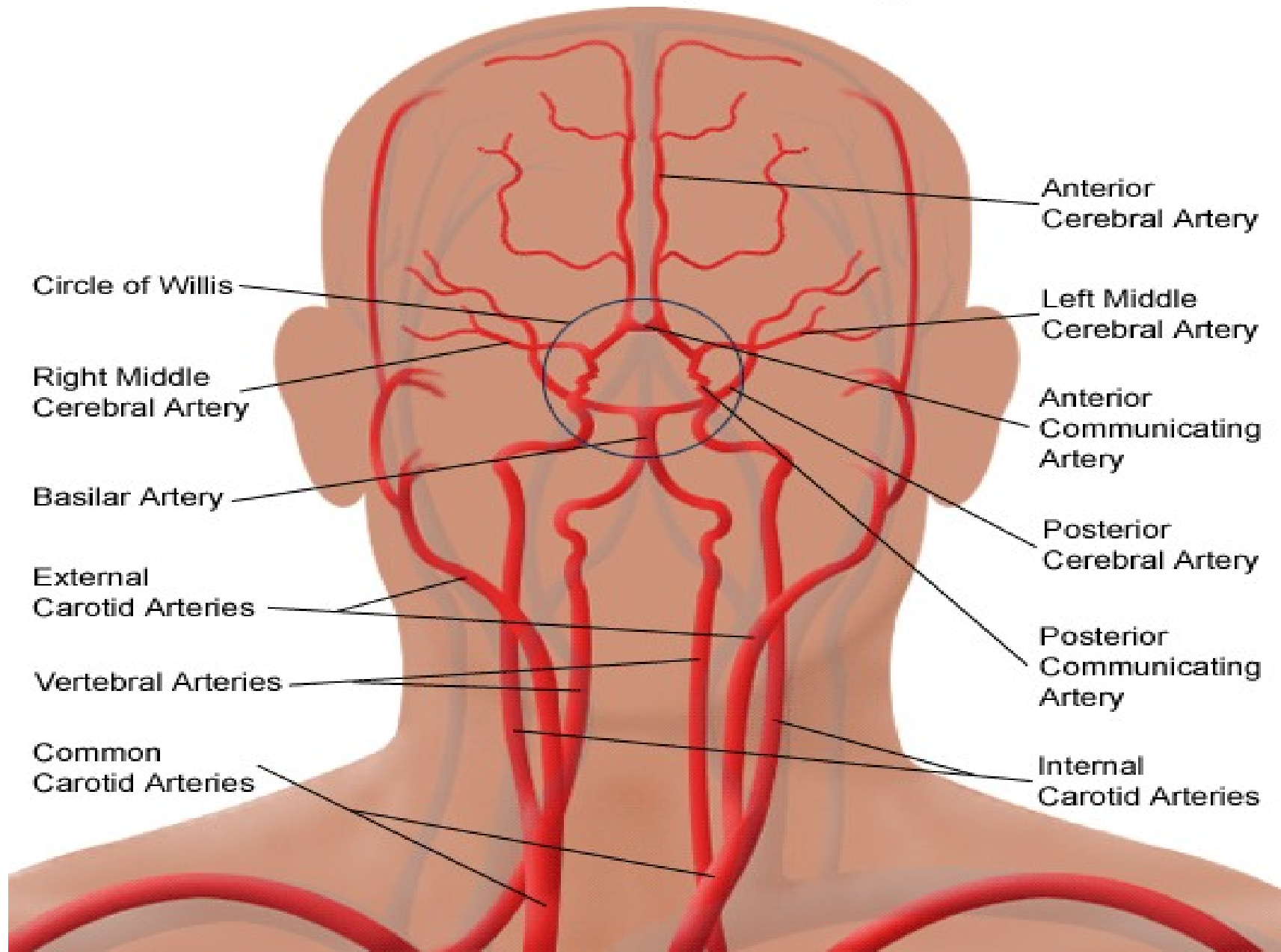


The brain is supplied by four arteries:

- **Vertebral arteries** (paired i.e right and left)
 - ✓ arises from the subclavian artery
 - ✓ ascends in the neck through the **transverse foramen**
 - ✓ enters the cranial cavity through the **foramen magnum**.
 - ✓ The right and left vertebral arteries unite to form the **basilar artery**.
- **Internal carotid arteries** (paired).

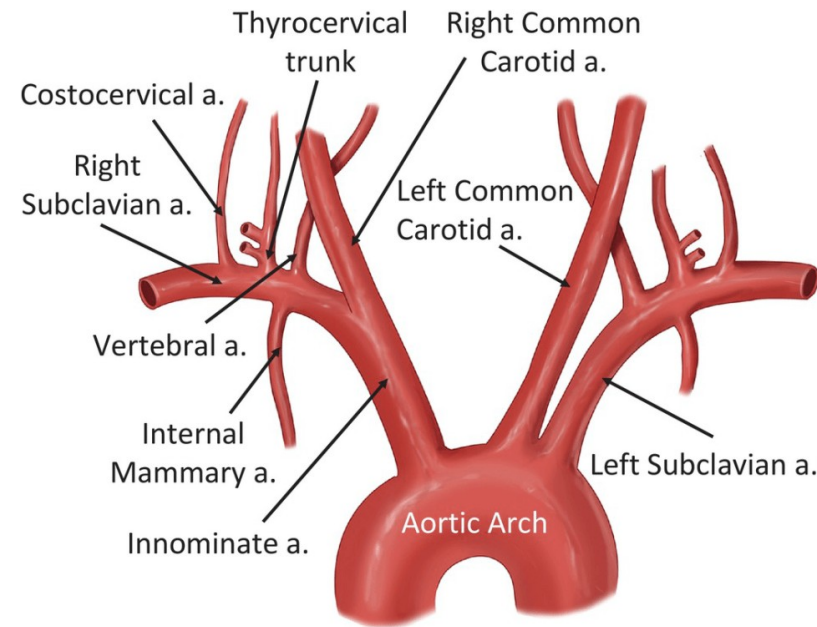


Arterial Circulation of the Brain, Including Carotid Arteries



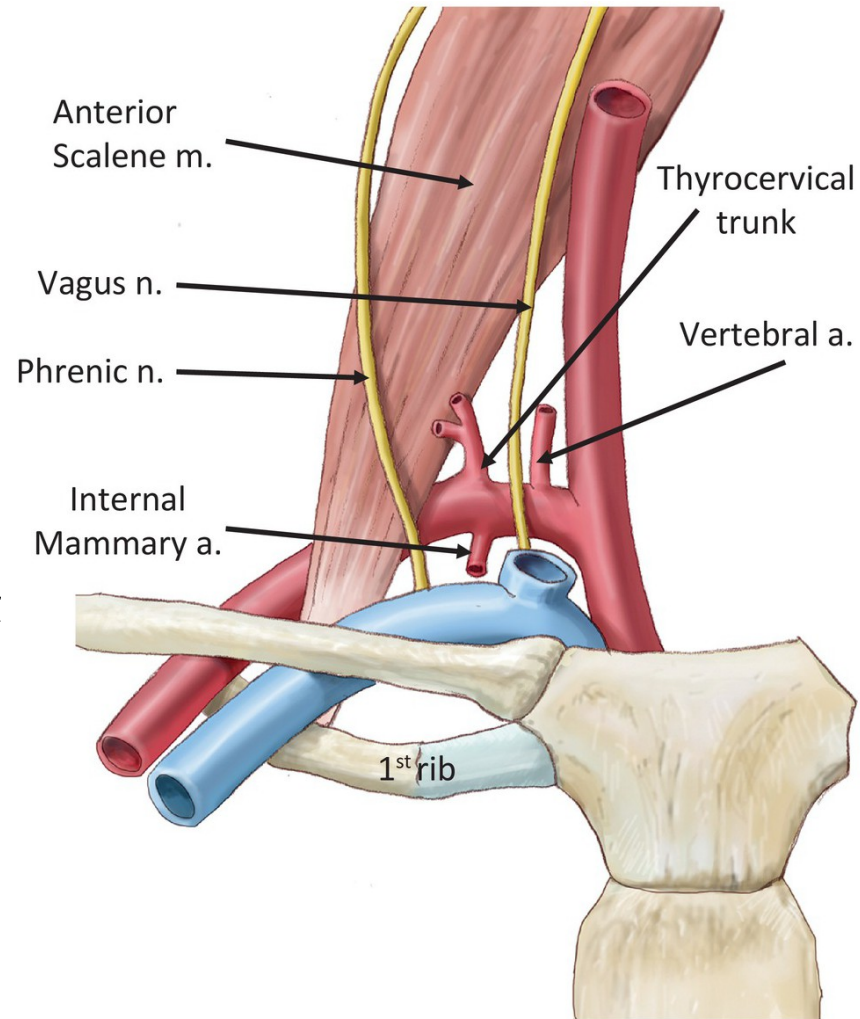
Subclavian arteries

- Supply the **upper limbs** and also send branches to the **neck** and **brain** .
- The **right** subclavian artery arises from the brachiocephalic trunk.
- The **left** subclavian artery arises from the arch of the aorta
- As the subclavian arteries cross the outer margin of the first ribs, their name changes to **axillary arteries**.



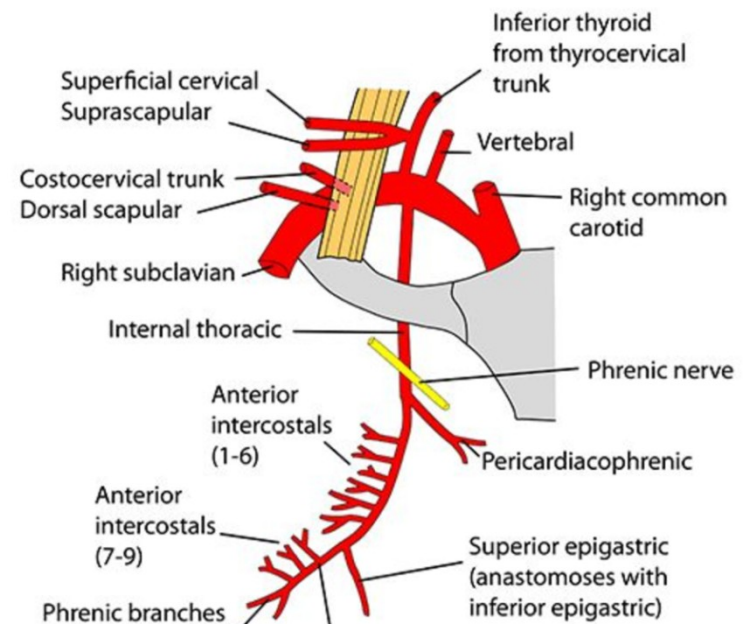
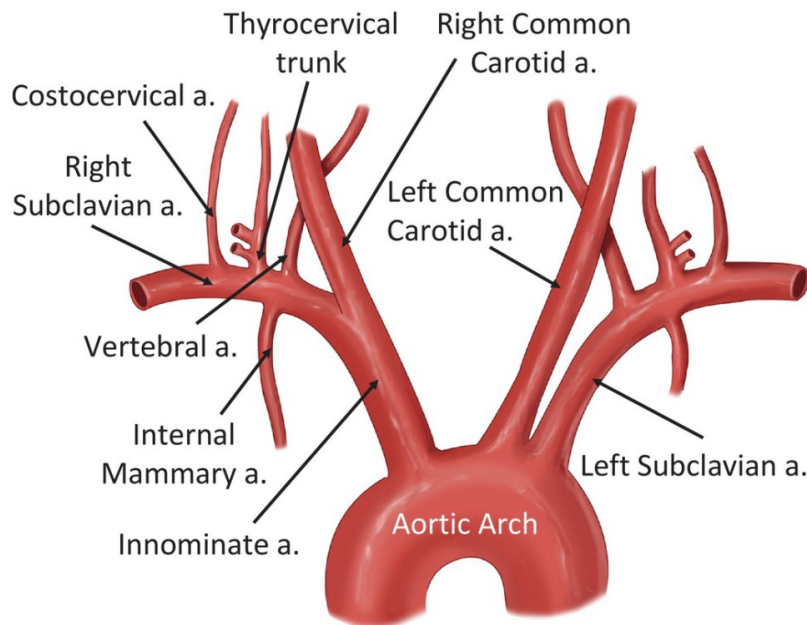
Subclavian arteries...

- Three parts of each subclavian artery are described relative to the **anterior scalene muscle** :
- the **first** part is medial to the it
- the **second** part is posterior to it
- the **third** part is lateral to it



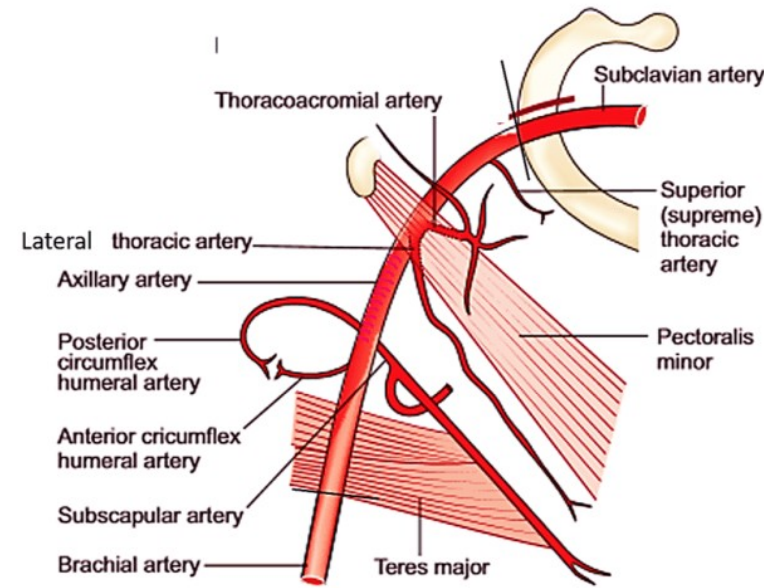
Branches of the subclavian arteries

- From the first part: **Vertebral** artery, **internal thoracic** artery, and **thyrocervical trunk**.
- From the second part: **Costocervical trunk**
- From the third part: **Dorsal scapular** artery,



Axillary artery

- Begins at the lateral border of the **1st rib** as the and ends at the inferior border of the **teres major**
- At inferior border of the teres major it becomes the **brachial artery**.
- For descriptive purposes, it is divided into three parts by the pectoralis minor

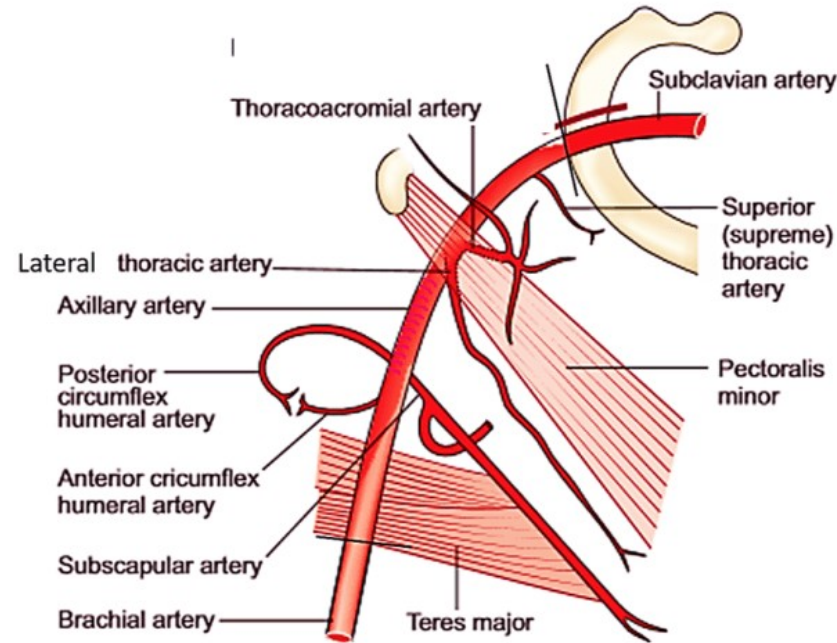


1. 1st part is b/n 1st rib and pectoralis minor and has **one branch**
✓ **Superior thoracic artery**
2. 2nd part lies posterior to pectoralis minor and has two branches **Thoracoacromial** and **Lateral thoracic**

Axillary artery...

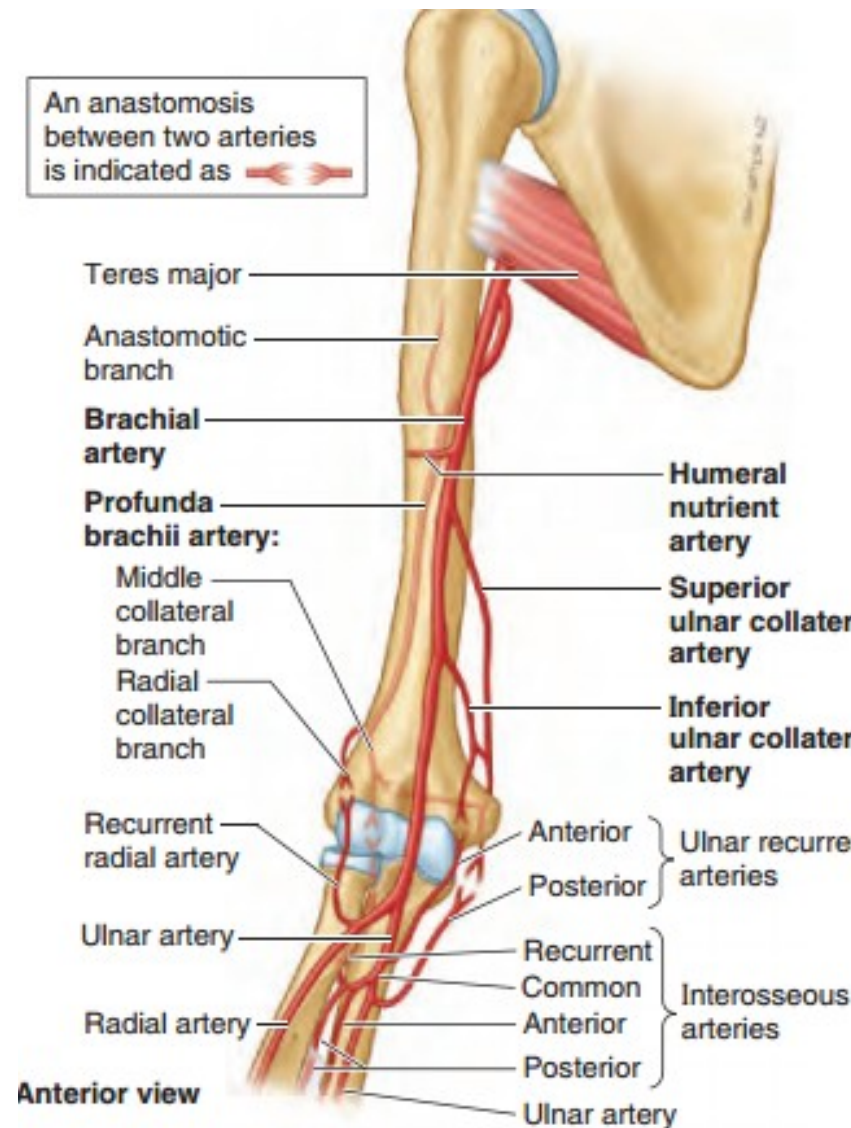
3. 3rd part extends from the lateral border of **pectoralis minor** to the **inferior border of teres major** and has three branches.

- ✓ **Subscapular artery,**
- ✓ **Anterior circumflex humeral**
- ✓ **Posterior circumflex humeral aa.**



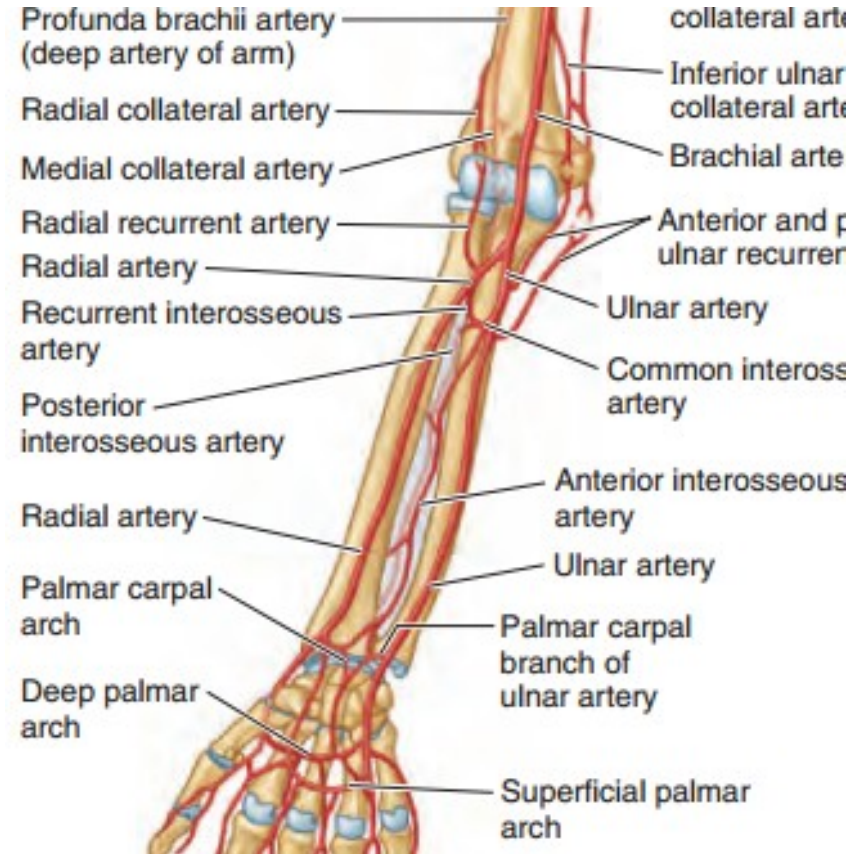
Brachial artery

- It begins at the inferior border of the teres major and ends in the cubital fossa
- The main named branches of the brachial artery are
 - Humeral nutrient artery
 - Profunda brachial artery
 - Superior ulnar collateral
 - Inferior ulnar collateral arteries
- In the cubital fossa, and brachial artery divides into the **radial** and **ulnar arteries**



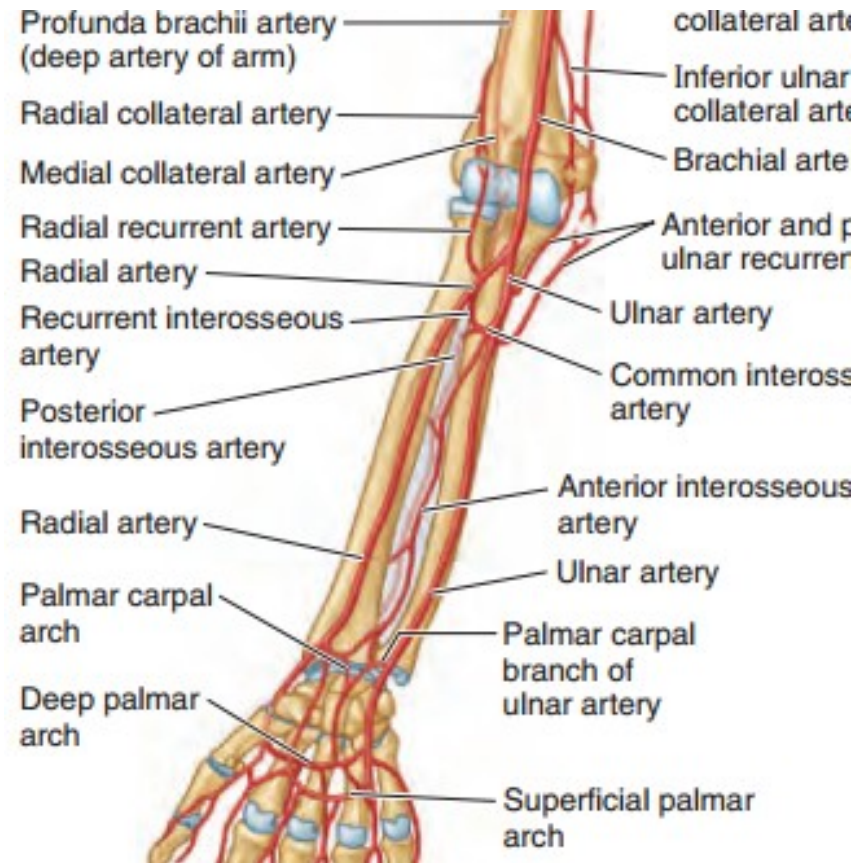
Radial artery

- It passes along the **lateral** aspect of the **forearm** then into the wrist and hand
- Because of its superficial location at wrist, it is a common site for measuring the **radial pulse**.
- Branches of radial artery
 - Radial recurrent artery
 - Palmar carpal branches of the radial artery
 - Dorsal carpal branches of the radial artery



Ulnar artery

- Passes along the **medial** aspect of the forearm and then into the wrist and hand.
- Branches of ulnar artery
 - *anterior ulnar recurrent artery*
 - *posterior ulnar recurrent artery*
 - *Common interosseous artery*
 - *Palmar carpal branches of the ulnar artery*
 - *Dorsal carpal branches of the ulnar artery*



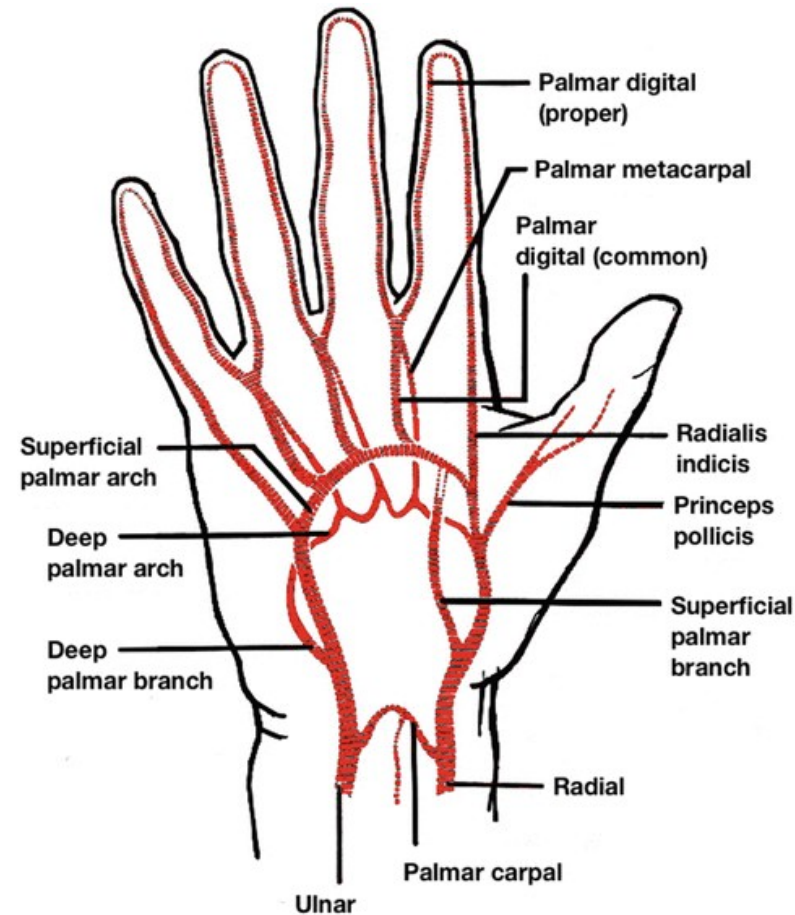
Arteries of the hand

- In the palm, branches of the radial and ulnar arteries anastomose to form the **superficial palmar arch** and the **deep palmar arch**.
- **Superficial palmar arch**
 - is formed mainly by the **ulnar artery**, with a branch of the **radial artery**.
 - It gives rise to **common palmar digital arteries**, which supply the palm.



- **Deep palmar arch**

- is formed mainly the **radial artery**, with branch of **ulnar aa**
- Arising from the deep palmar arch are **palmar metacarpal arteries**, which supply the palm and anastomose with the common palmar digital arteries

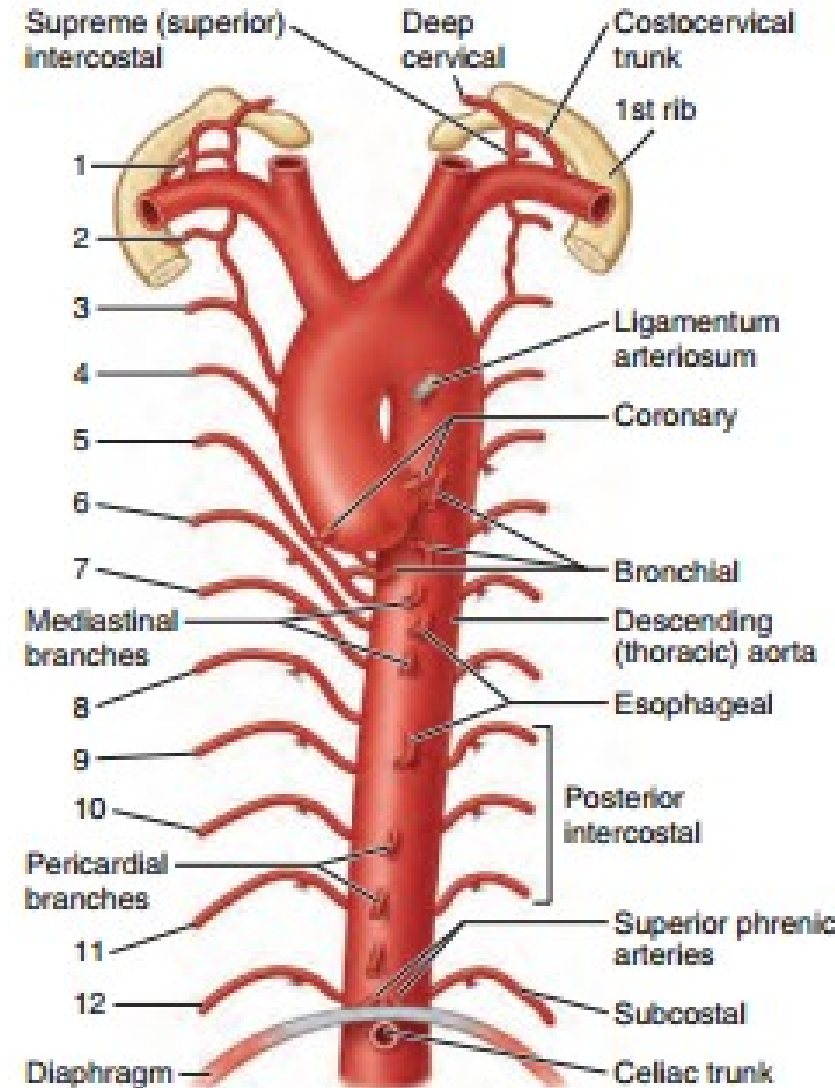


Thoracic aorta

- The is the continuation of the arch of the aorta
- It begins at the level of the inferior border of the body of the **T4 vertebra**
- It terminates (with a name change to *abdominal aorta*) anterior to the inferior border of the T12 vertebra and enters the abdomen through the **aortic hiatus** in the diaphragm

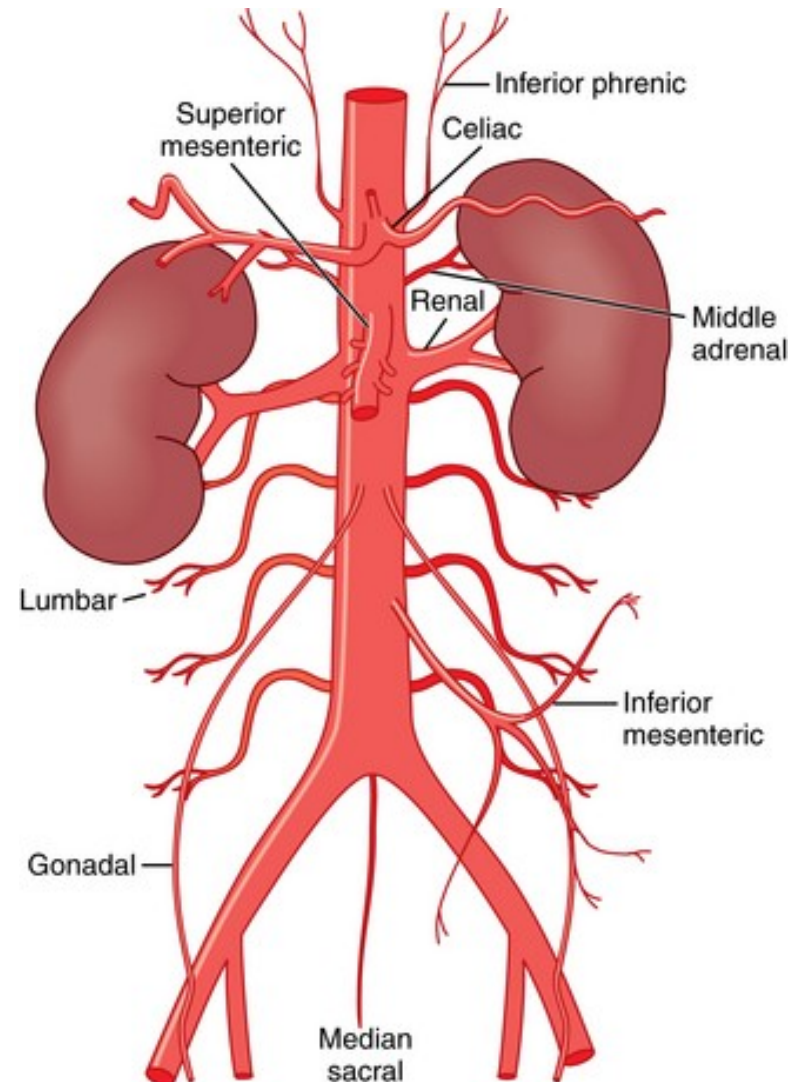
Thoracic aorta...

- Branches of the thoracic aorta categorized as
 - **Paired** and **unpaired** branches
- **Unpaired branches**
 - Esophageal
 - Pericardial
 - Mediastinal
- **Paired branches**
 - Bronchial
 - Posterior intercostals
 - Subcostal
 - Superior phrenic



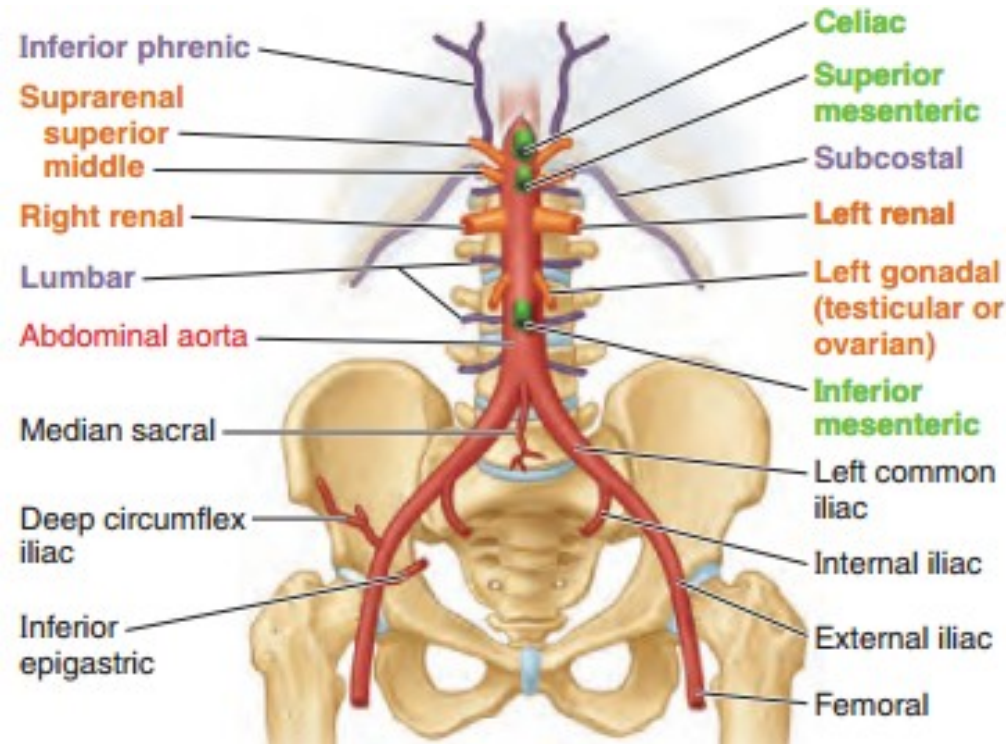
Abdominal aorta

- It **begins** at the aortic hiatus in the diaphragm at the level of the **T12 vertebra** and **ends** at the level of the **L4 vertebra** by dividing into the **right** and **left common iliac arteries**.



Branches of abdominal aorta

- Abdominal aorta has paired branches and unpaired branches
- **Unpaired branches**
 - Celiac trunk
 - Superior mesenteric artery
 - Inferior mesenteric artery
 - Median sacral artery
- **Paired branches**
 - Inferior phrenic aa.
 - Middle suprarenal aa.
 - Renal aa.
 - Gonadal (testicular or ovarian) aa.
 - Lumbar aa.



- **Celiac trunk**
 - Supplies all the structures related to **abdominal foregut**
- **Superior mesenteric artery**
 - Supplies structures of **abdominal midgut**
- **Inferior mesenteric artery**
 - Supplies structures of **abdominal hindgut**

Divisions of the Digestive Tract

Foregut = esophagus (abdominal) to D3/D4 segment (1-2-3-4)

Small Intestine

Duodenum

Jejunum and Ileum

Ascending Colon

Midgut = D4, jejunum to (rest of small bowel)
colon (ascending and transverse colon)

Rectum

everything else

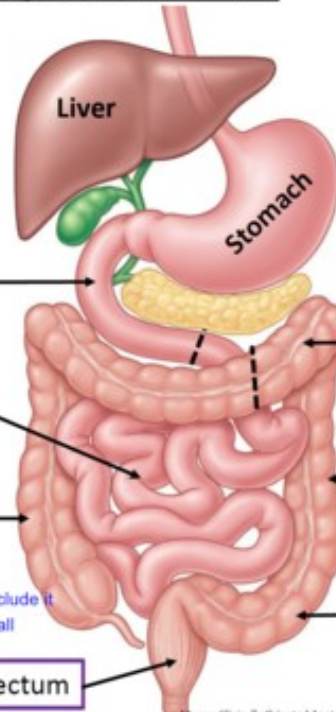
Hindgut = distal 1/3 transverse colon to rectum

Large Intestine (colon)

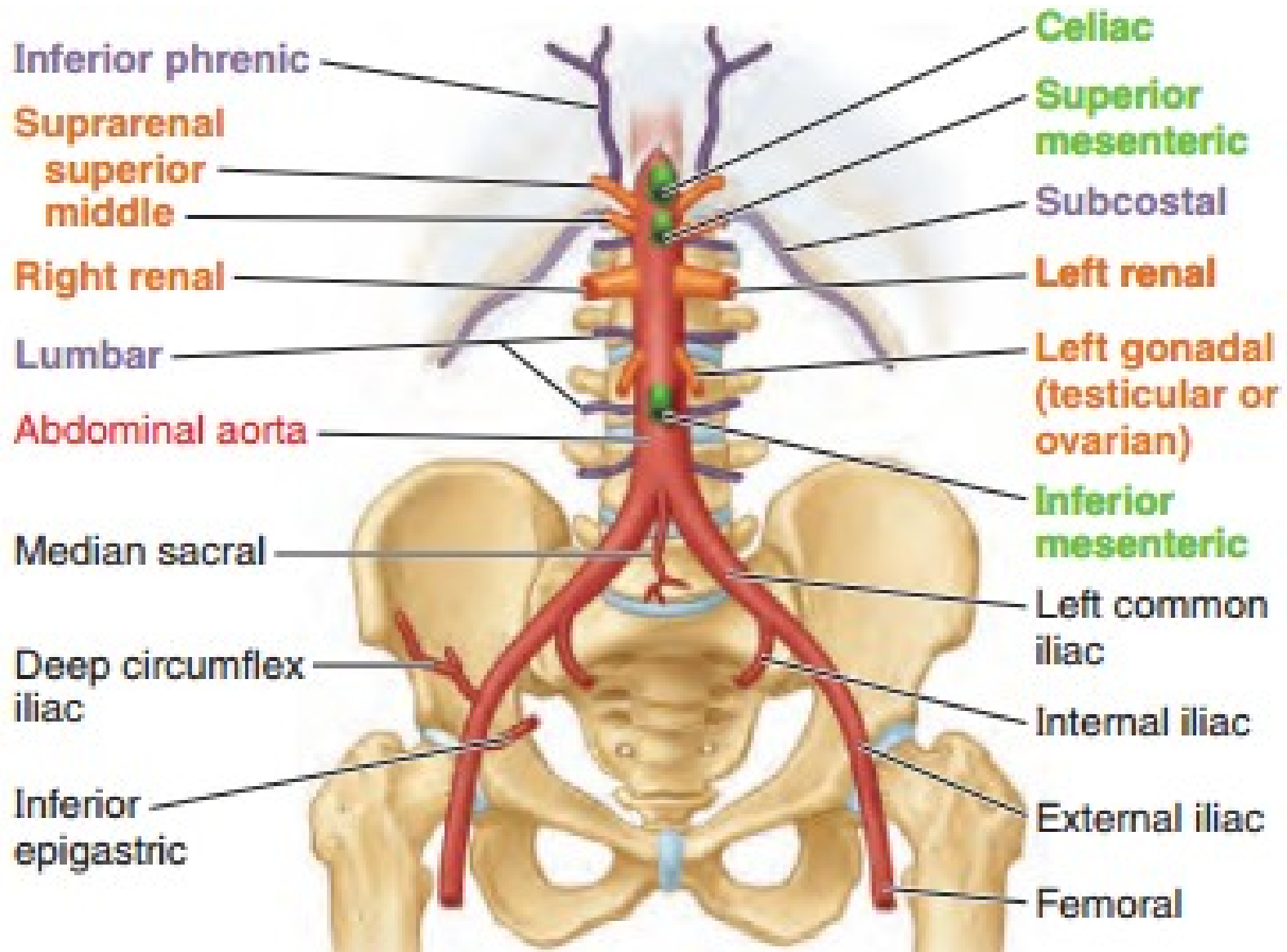
Transverse Colon

Descending Colon

Sigmoid Colon

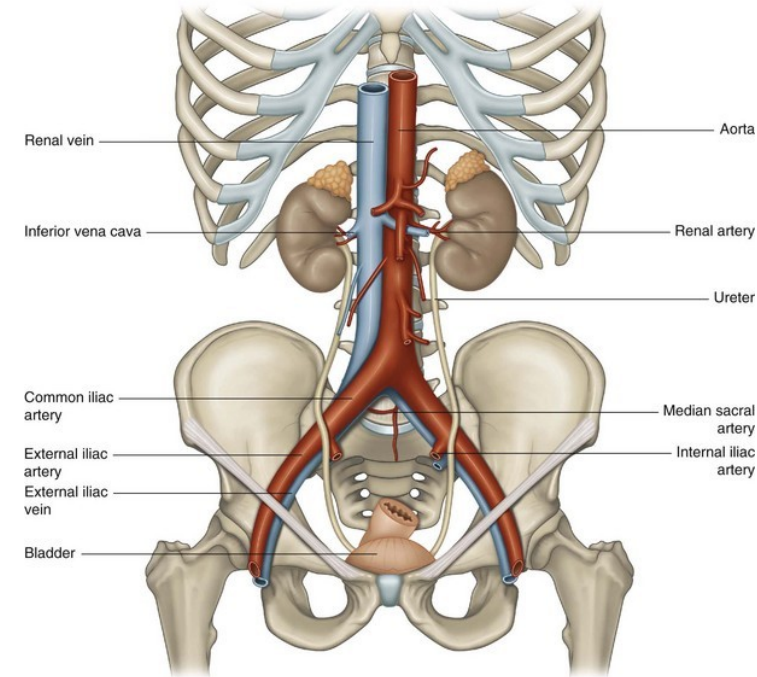


Abdominal aorta



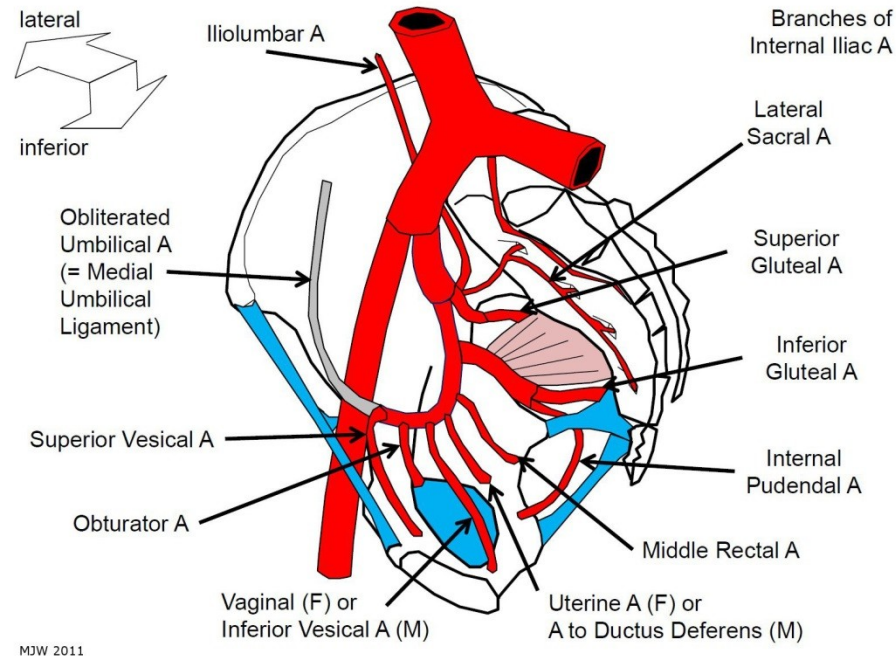
Arteries of the pelvis and lower limbs_

- The abdominal aorta terminates by bifurcating into
 - Right common iliac arteries
 - Left common iliac arteries
- The common iliac divides into
 - Internal iliac
 - External iliac.



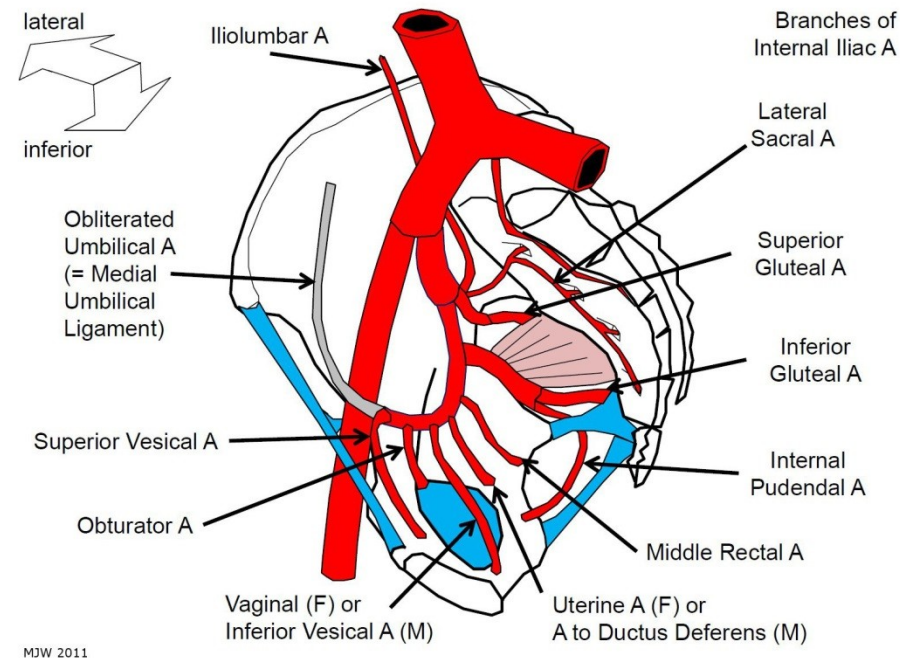
Internal iliac (hypogastric) arteries

- are the primary arteries of the **pelvis**.
- as they descend in the pelvis and divide into **anterior and posterior divisions**.
- The general distribution of the internal iliac arteries is to the **pelvis, buttocks, external genitals, and thigh**.



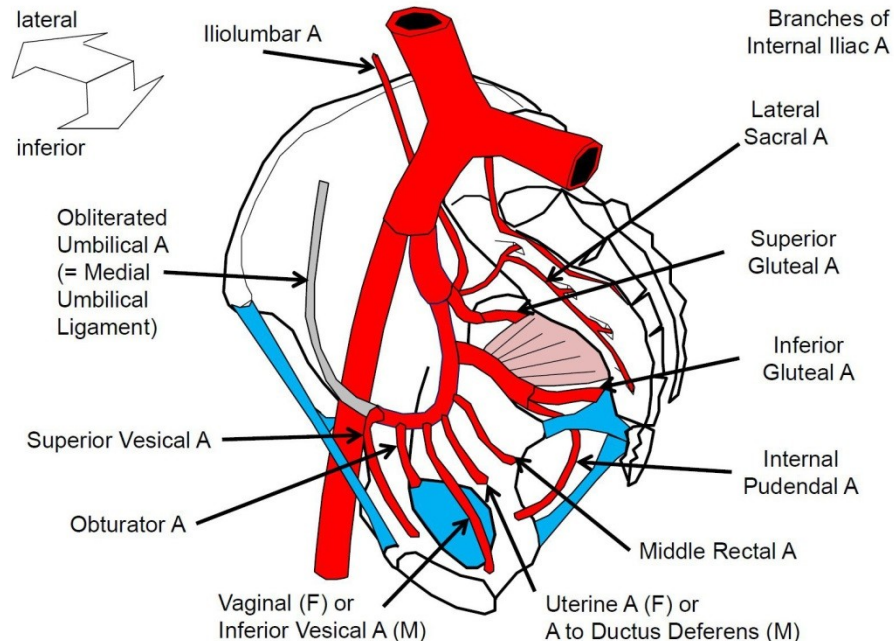
Internal iliac arteries...

- The branches of the **anterior division** are
 - Umbilical Artery
 - Obturator Artery
 - Inferior Vesical/Vaginal Artery
 - Uterine Artery/Ductus Deference
- The branches of the **posterior division** of the are
 - Lateral sacral arteries:
 - Superior gluteal artery:
 - Iliolumbar artery



External iliac arteries

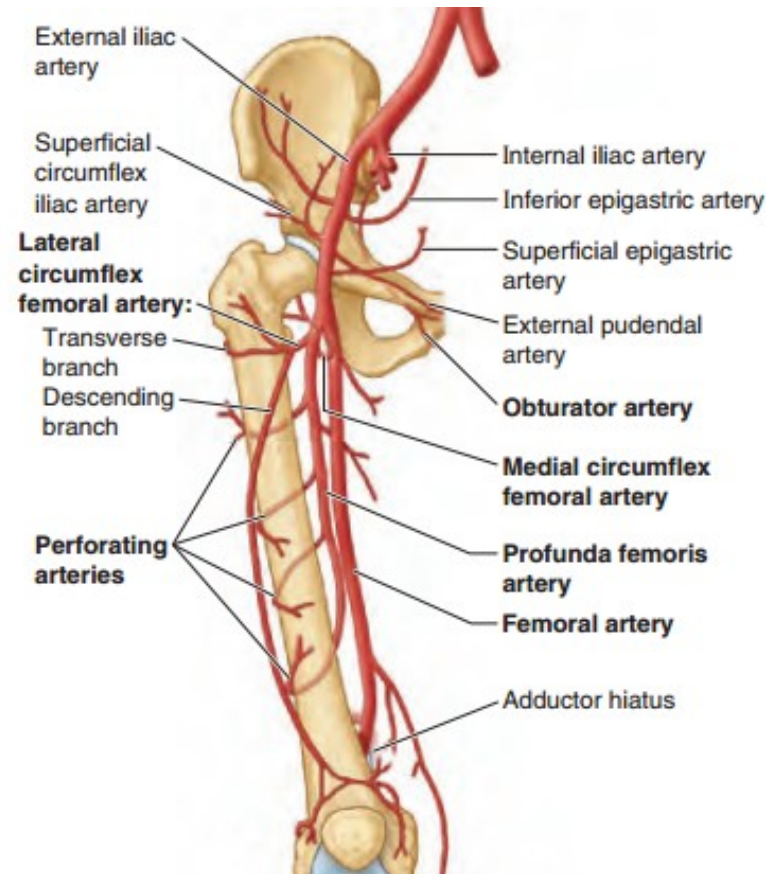
- They are larger than the internal iliac arteries.
- pass posterior to the inguinal ligaments, and become the **femoral arteries**.
- The general distribution of the external iliac arteries is to the **lower limbs**.



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Femoral artery

- Is the continuation of the external iliac artery distal to the **inguinal ligament**
- is the primary artery of the lower limb
- A major branch of the femoral artery:
 - Superficial epigastric aa.
 - Superficial circumflex iliac aa.
 - Superficial external pudendal aa.
 - Deep external pudendal aa.
 - Profunda femoris aa.

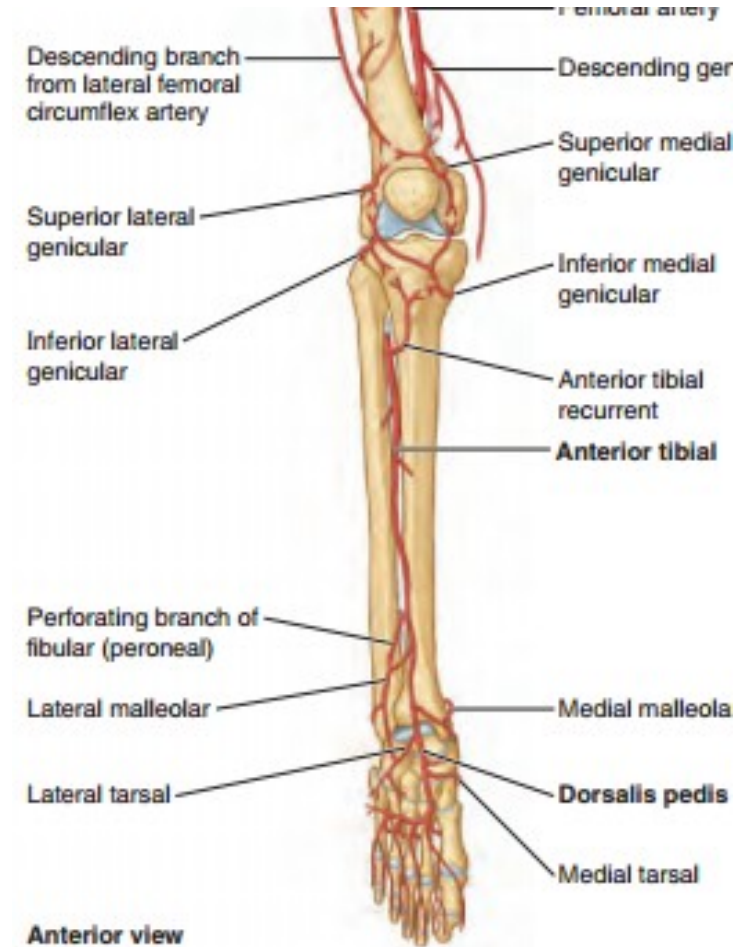


Popliteal artery

- Femoral artery becomes the popliteal artery as it passes across the posterior aspect of the knee.
- Passes infero laterally through the ends at the **inferior border of the popliteus** by dividing into the **anterior** and **posterior tibial arteries**.
- Five **genicular branches** of the popliteal artery supply the **capsule and ligaments of the knee joint**.

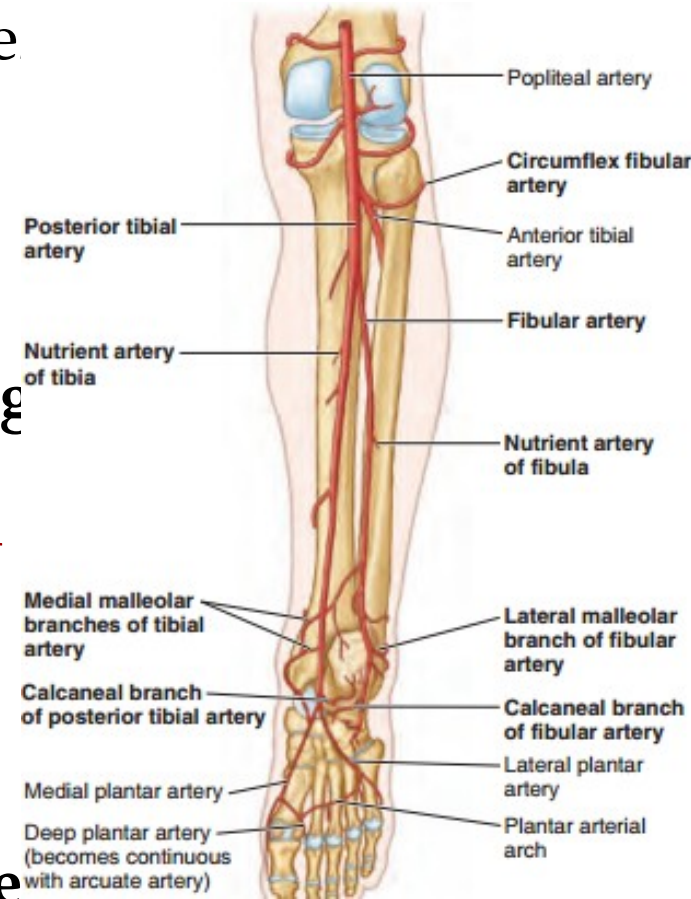
Anterior tibial arteries

- Pass anteriorly on the opening above interosseous membrane
- at the ankle, becomes the **dorsal pedal artery**
- The **dorsal pedal artery** is the site from which the most distal pulse is recorded.
- Generally **Anterior tibial arteries** supply anterior compartment of the leg and muscle at the dorsum of the foot



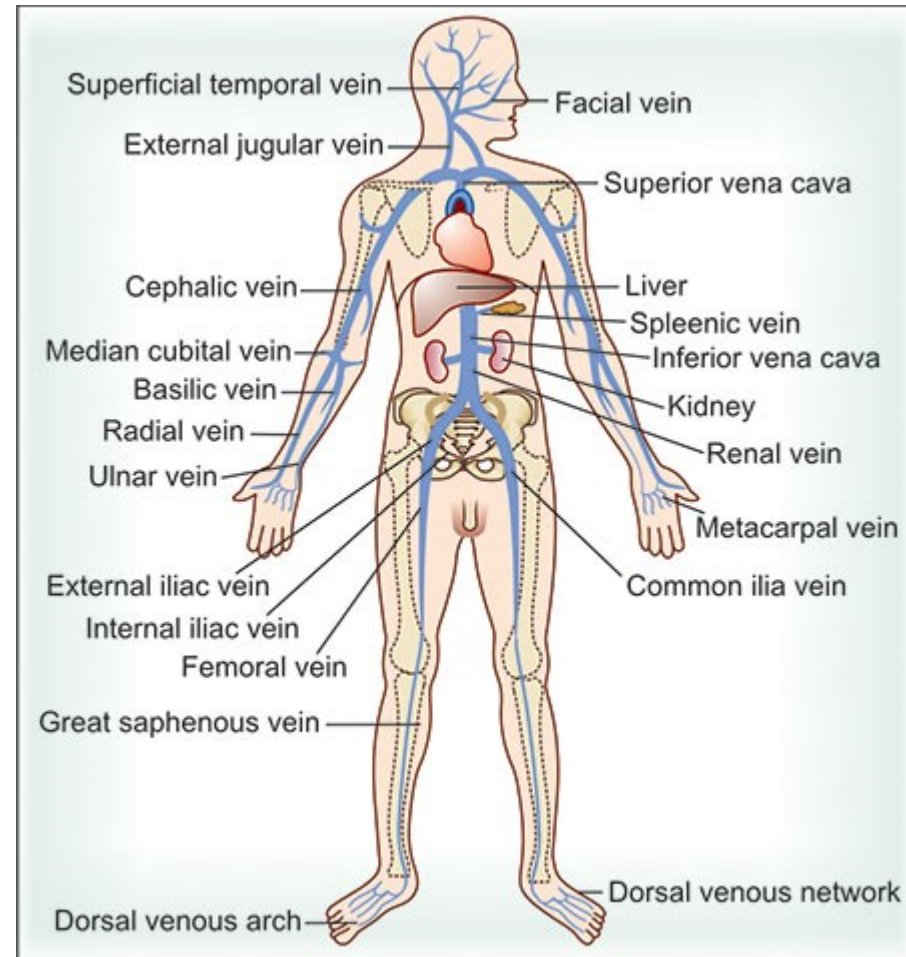
Posterior tibial arteries

- Is the largest of the two terminal branches of popliteal artery
- forms the **large fibular artery** which supplies the **peroneal muscles** of the leg
- At the ankle it bifurcates into the **lateral** and **medial plantar** arteries.
- The **lateral plantar artery** forms the **plantar arch** and gives off **digital arteries** to the digits of the foot.
- Generally **Posterior tibial arteries** supply the **posterior** and **lateral** compartment of the **leg** as well as the **sole of the foot**.



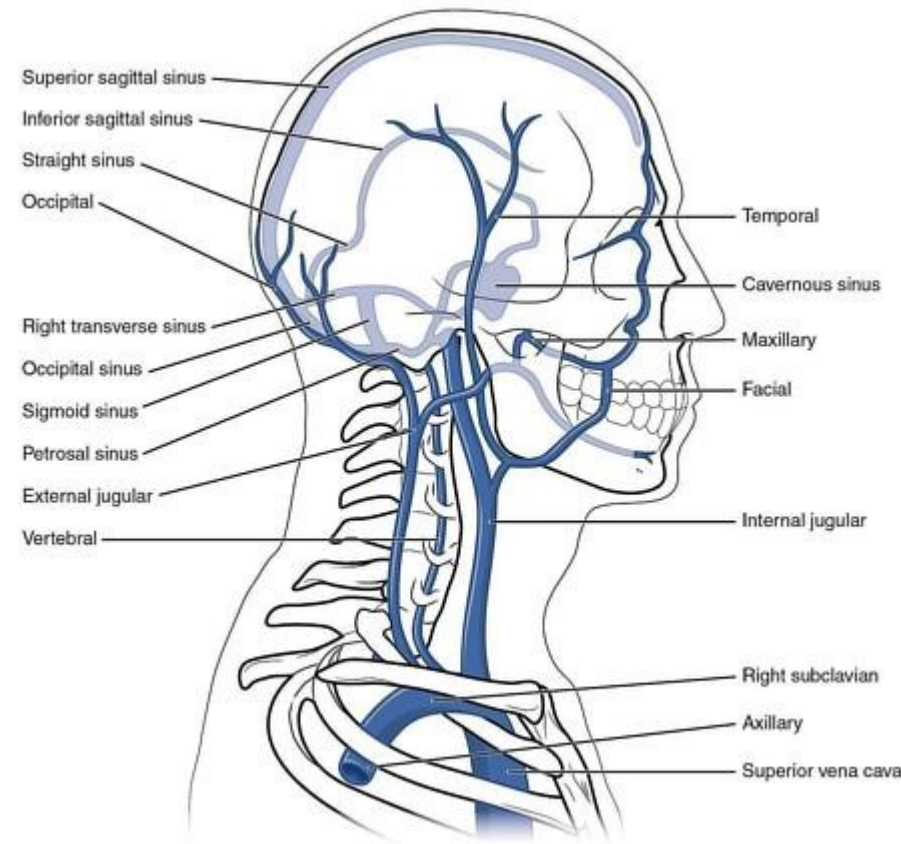
Venous drainage of the body

- Veins Draining the neck & head
- Veins of the upper extremity
- Veins of the thorax
- Veins of the abdomen
- Portal system
- Veins of the lower extremity



Internal jugular veins

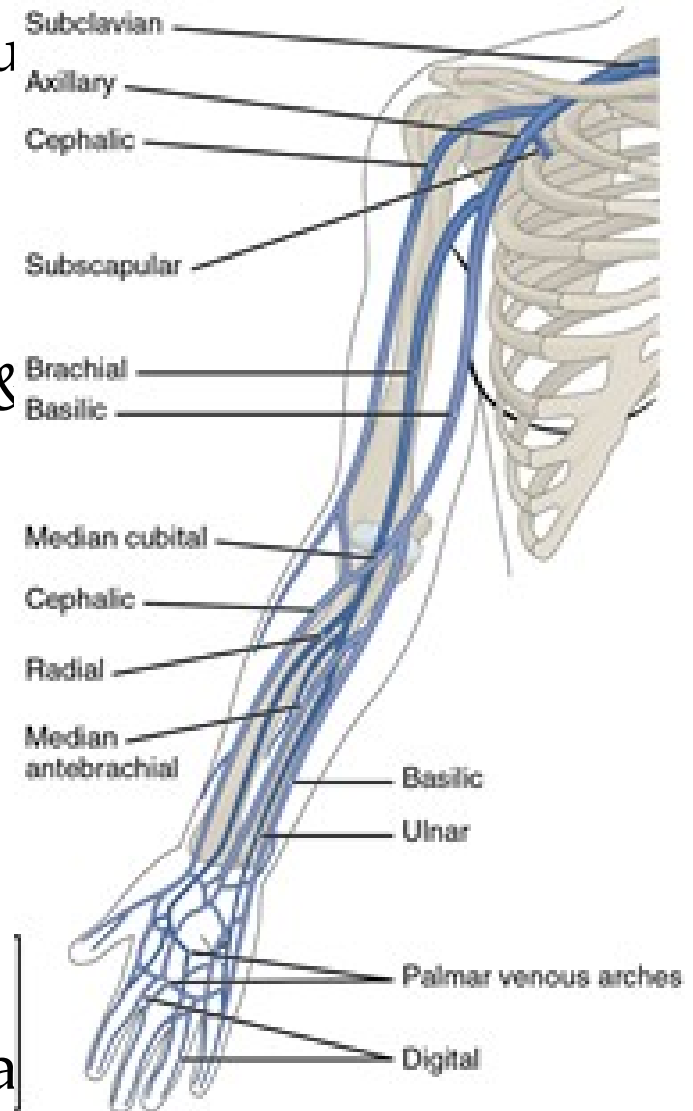
- The right and left internal jugular veins pass inferiorly on either side of the neck
- from brain, meninges, deep regions of face and neck
- then unite with the **subclavian veins** to form the right and left brachiocephalic veins
- From here blood flows into the superior vena cava.



- **External jugular vein**
 - ✓ from scalp, portions of face, superficial neck region
 - ✓ drain into right and left subclavian vein.

Veins of the upper extremity

- Consists of superficial and deep venous drainage
- **Deep veins of UL**
 - accompany the arteries and bear their names / region- **radial vein & ulnar vein**
 - both drain from **deep** and **superficial** palmar arches
 - radial and ulnar veins join in the cubital fossa to form the **brachial vein**, which continues up on the medial side of the arm.
 - Brachial vein -axillary → subclavia → brachiocephalic → SVC



Veins of the upper extremity...

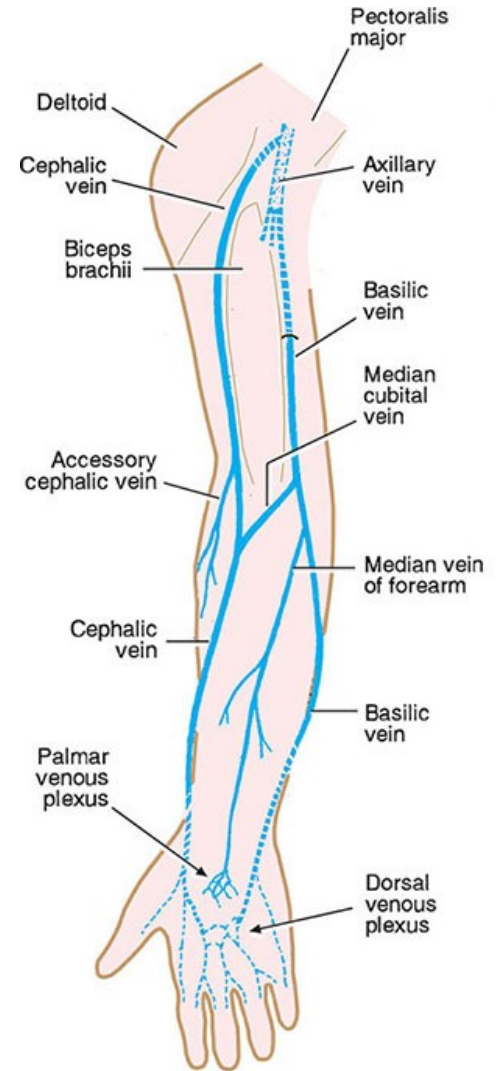
- **Superficial veins of UL**

- 1. **Basilic vein**

- drains blood from **medial** side of **forearm** and **arm**
 - merges with brachial vein near the head of the humerus to form the **axillary vein**

- 2. **Cephalic vein**

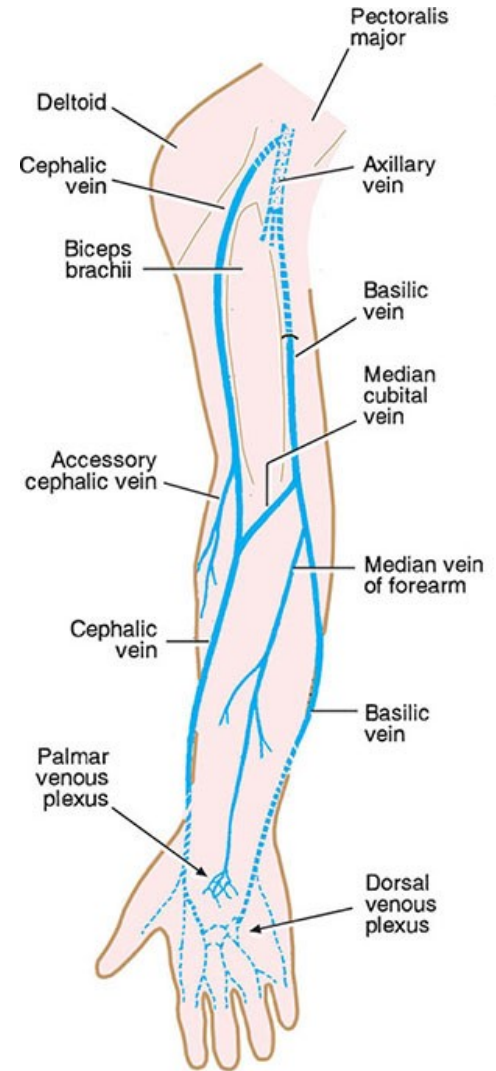
- drains superficial region of hand and forearm on radial side
 - Joins axillary vein in the shoulder region



Superficial veins of UL...

3. Median antebrachial vein (median vein of the forearm)

- Drains from palmar venous plexus and ascends in the middle of the anterior aspect of the forearm between the cephalic and the basilic veins and sometimes joins the **basilic** or **cephalic** or **median cubital vein**
- **Median cubital vein** ascends from the cephalic vein obliquely to join basilic vein in the cubital fossa.
- It is a site of **venipuncture**



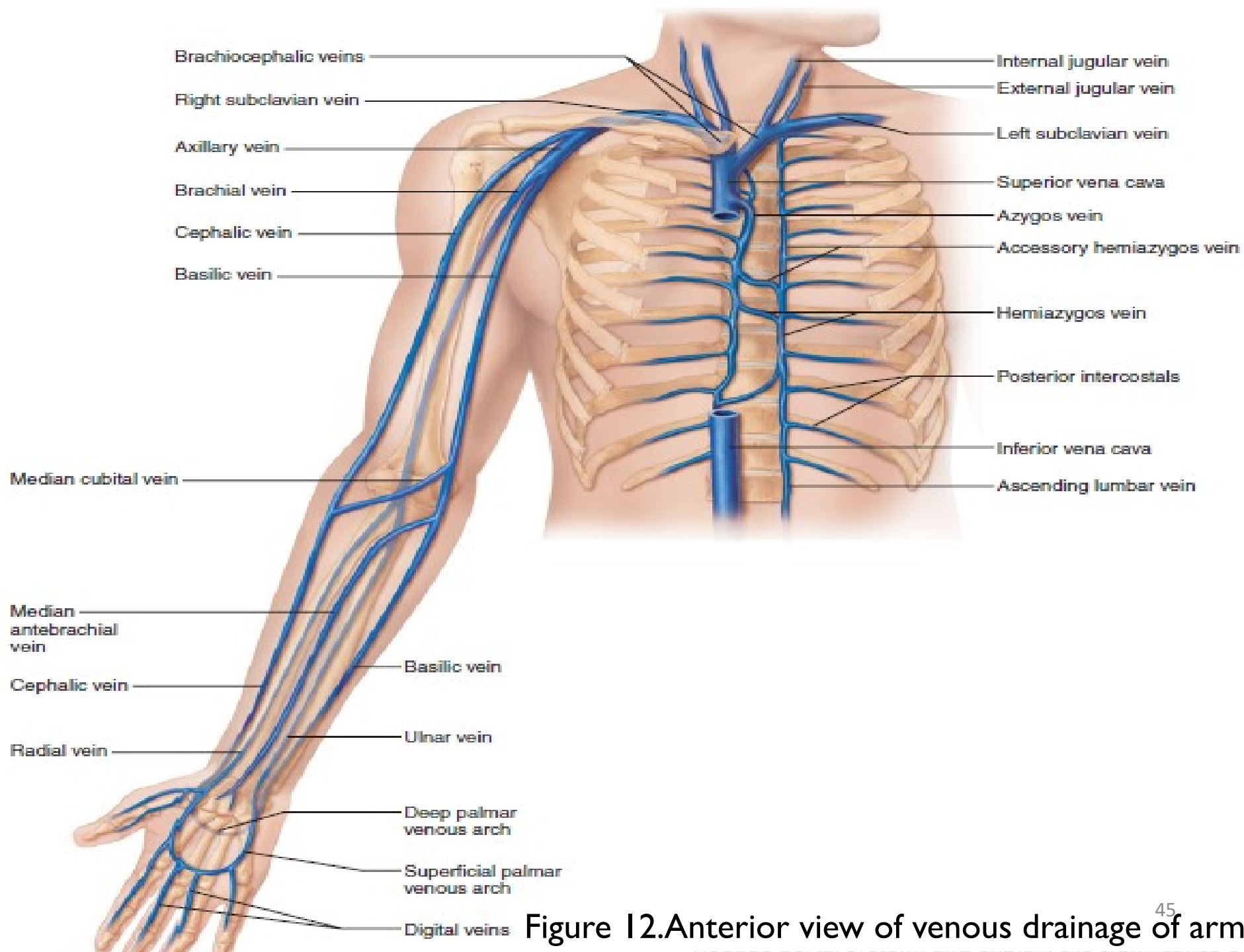
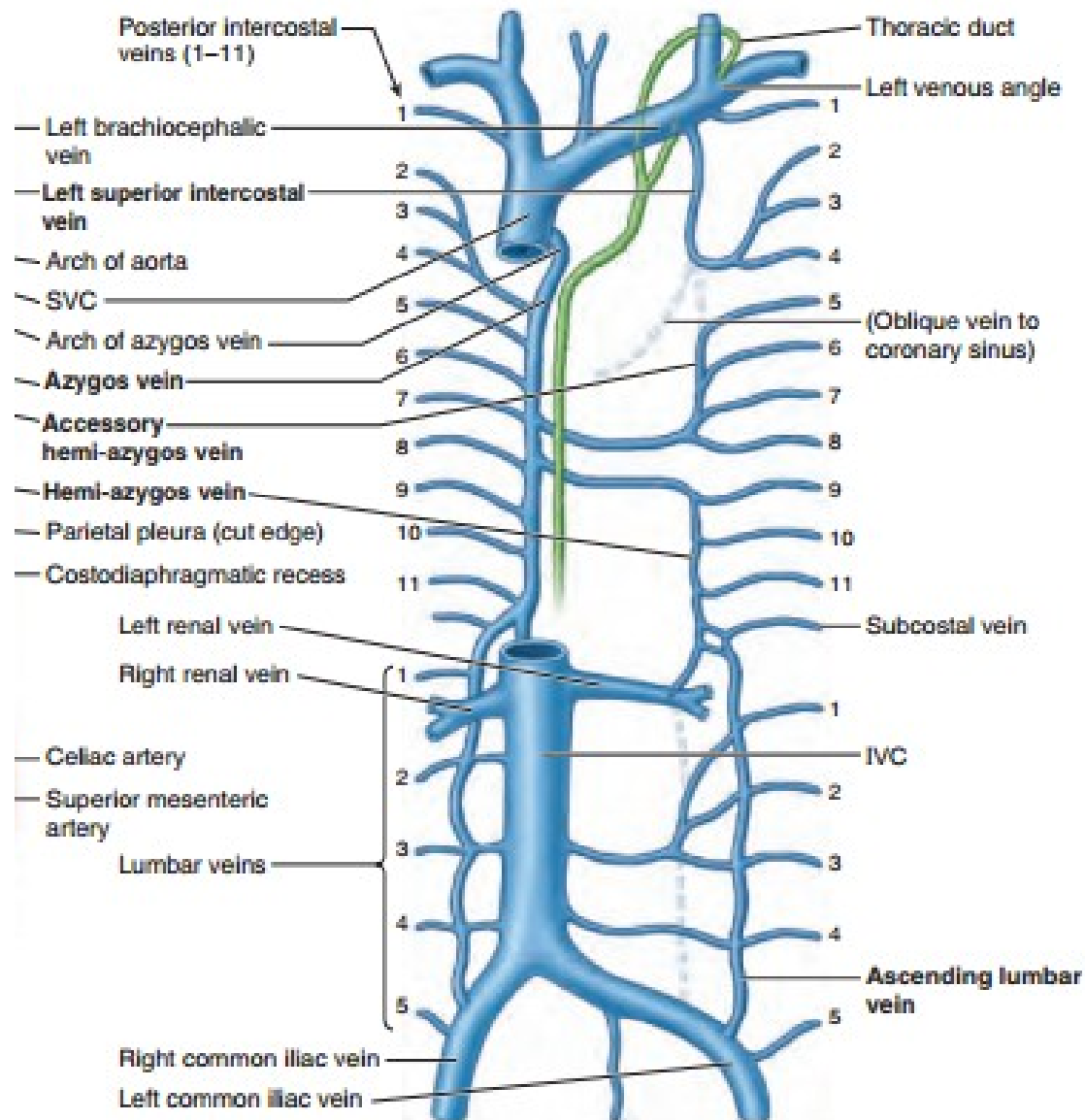


Figure 12. Anterior view of venous drainage of arm

Veins of the thorax

- Superior vena cava
 - receives blood from the right and left **brachiocephalic veins**, which drain **head, neck, and upper limb** as well as from **azygous veins**.
 - lacks valves which are characteristics of most veins
- The azygous vein
 - extends superiorly along the dorsal abdominal and thoracic walls on the right side of the vertebral column
 - joins the superior vena cava at T4.
 - Its tributaries are:
 - **Ascending lumbar veins** - drain from lumbar and sacral regions
 - **Intercostal veins**- from intercostals regions
 - **Accessory hemiazygous** and **hemiazygous veins**– left side of the thoracic wall, thoracic viscera, and abdominal wall.

Azygous vein...



Veins of the lower extremity

Deep vein

- **Anterior tibial veins**

- Drains from dorsal venous arch joins fibular and posterior tibial vein and ascend upward and form the **popliteal vein** to the back of the knee.

- **Posterior tibial and fibular veins**

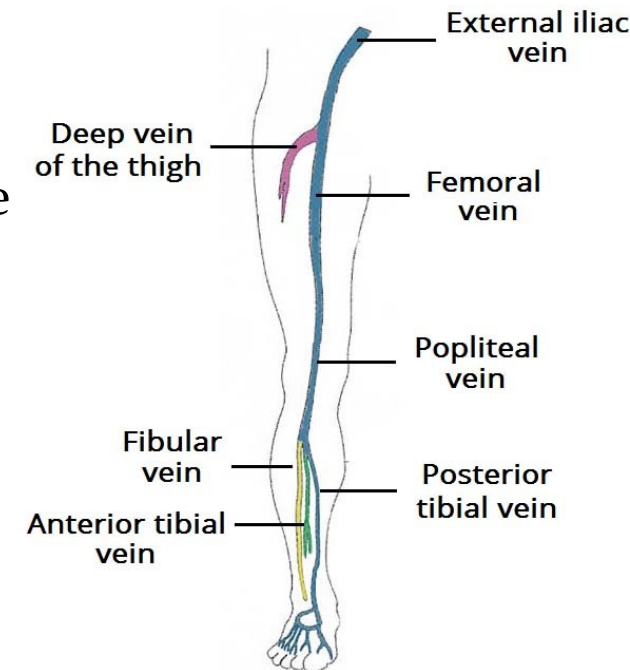
- Medial and lateral plantar vein from the plantar aspect of the foot form posterior tibial and fibular vein posterior to the medial and lateral malleoli

- **Popliteal vein**

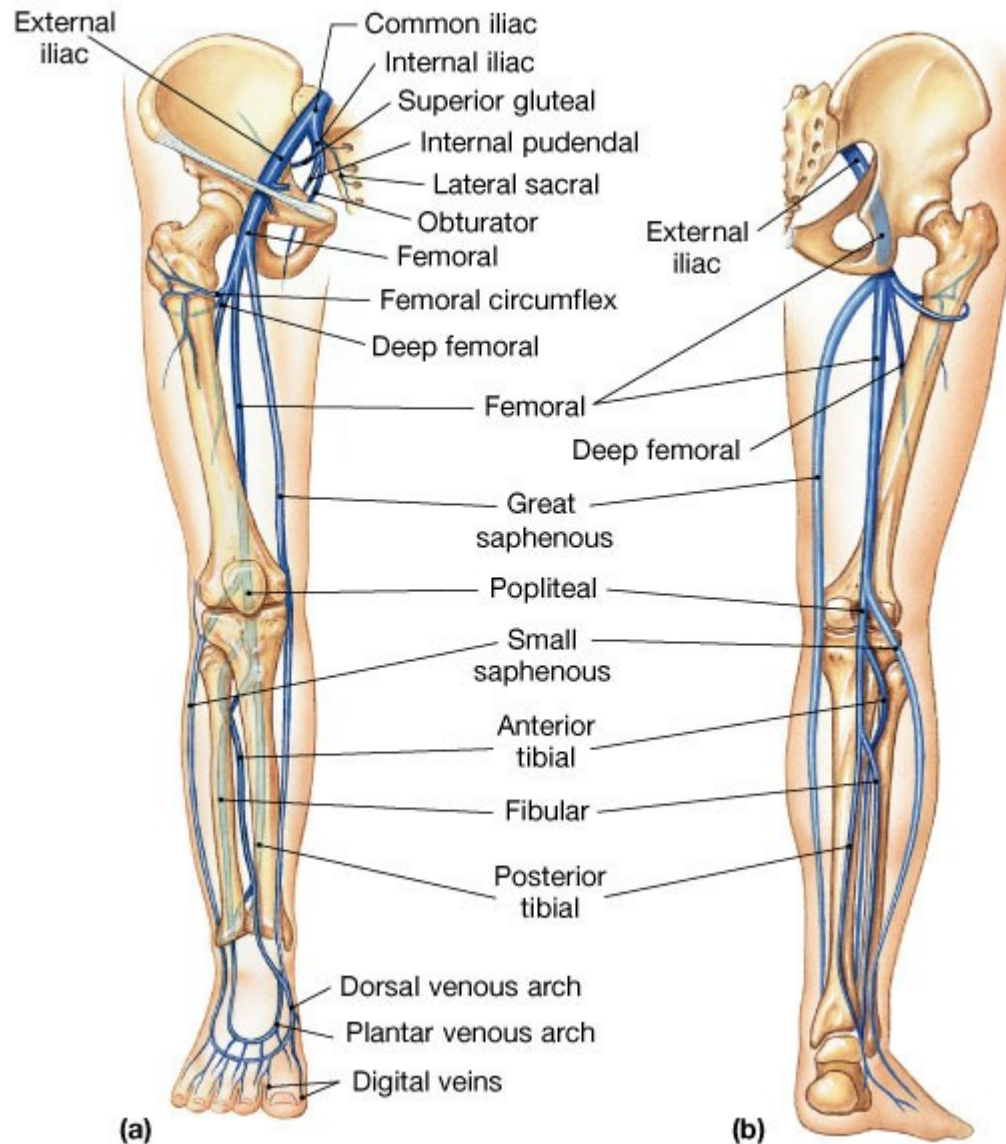
- drains blood from the knee region and above the knee, it becomes the femoral vein

- **Femoral vein**

- Receives blood from the **deep vein of the thigh** and **great saphenous vein**, then becomes the **external iliac vein** as it passes under the inguinal ligament.



Veins of the lower extremity...



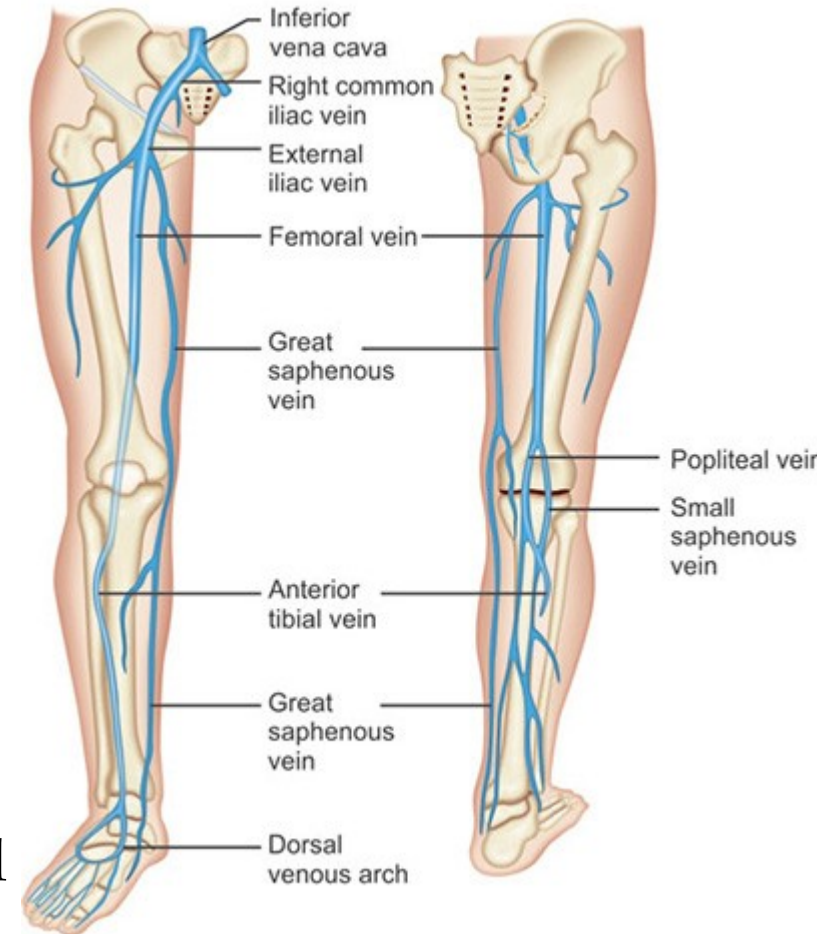
❖ Superficial vein include

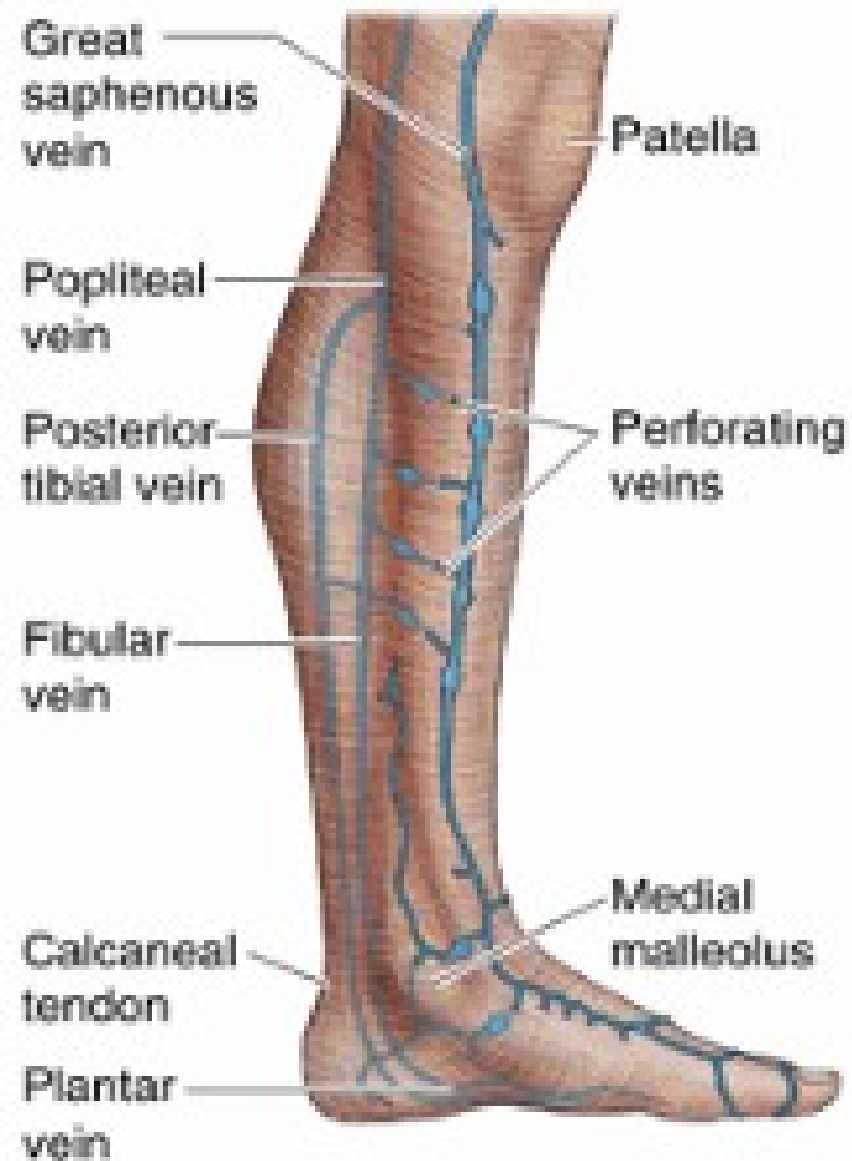
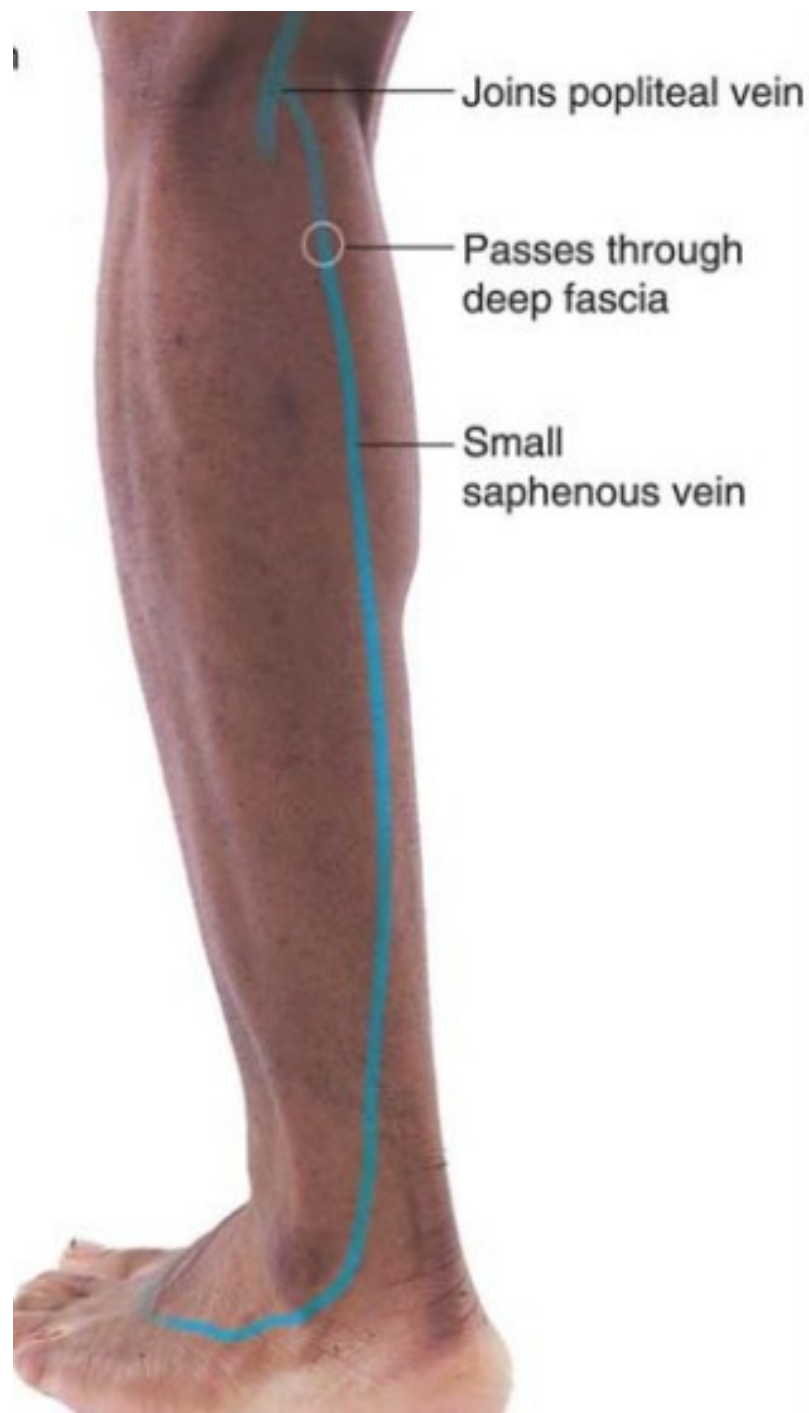
■ Small saphenous vein

- begin at the **lateral** aspect of the **dorsal venous arches** of the foot
- ascend deep to the skin along the posterior aspect of the leg.
- they empty into the popliteal veins in the **popliteal fossa**

■ Great saphenous vein

- longest vessel in the body
- originates at the arch of the foot and ascends along the **medial** aspect of the **leg** and **thigh** then draining into the **femoral vein**.



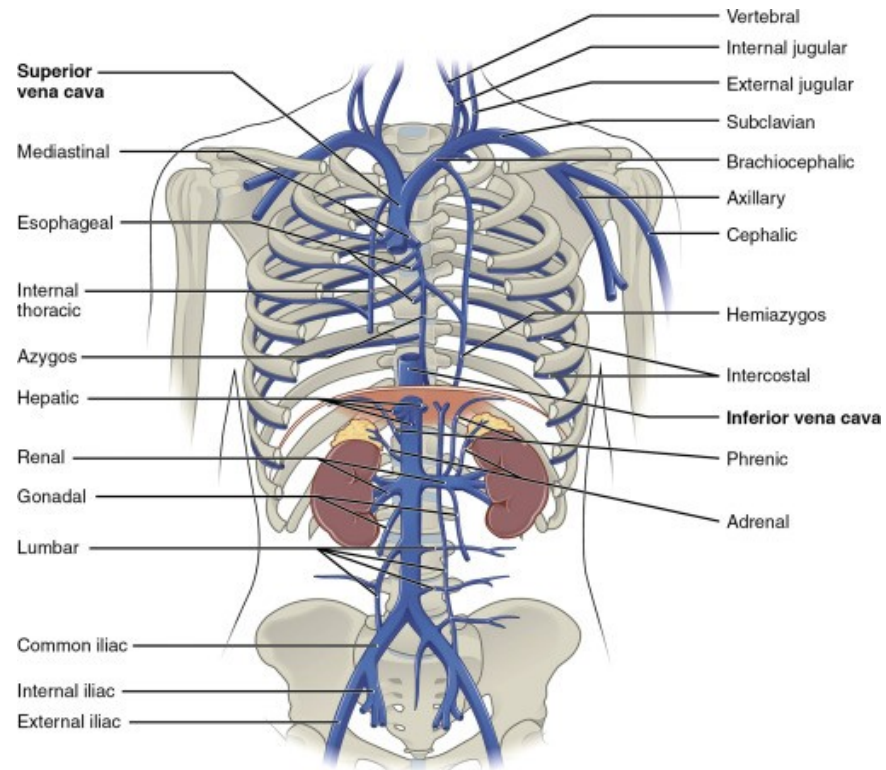


(D) Medial view

Veins of the Abdominal Region

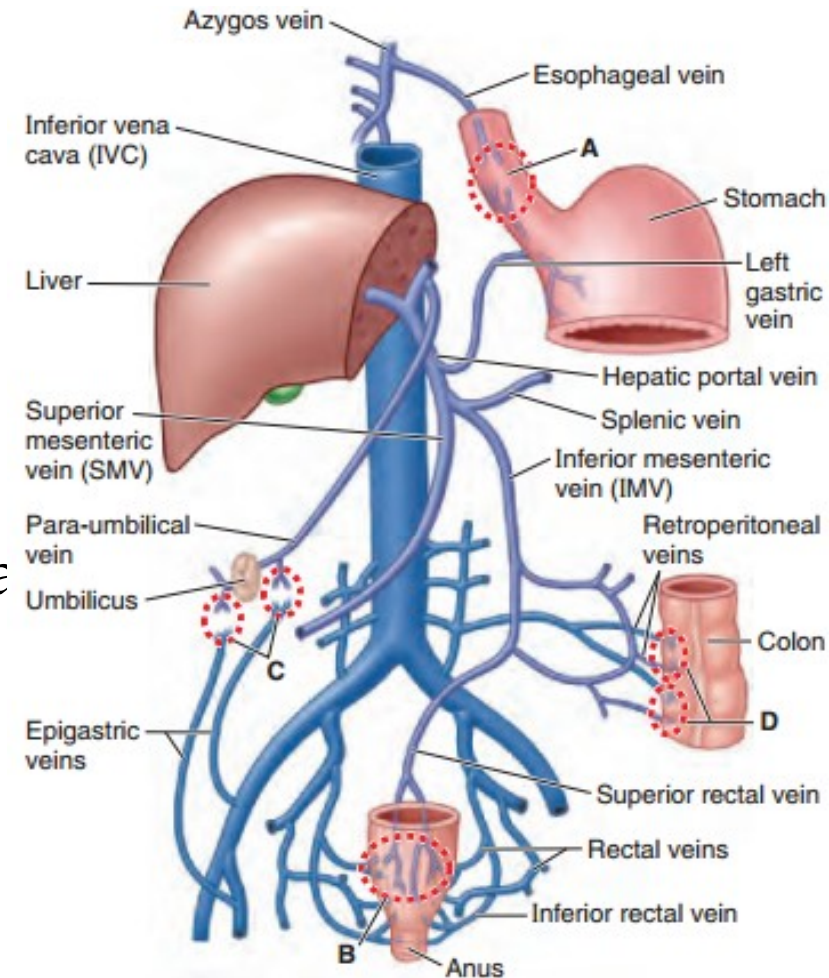
Inferior vena cava

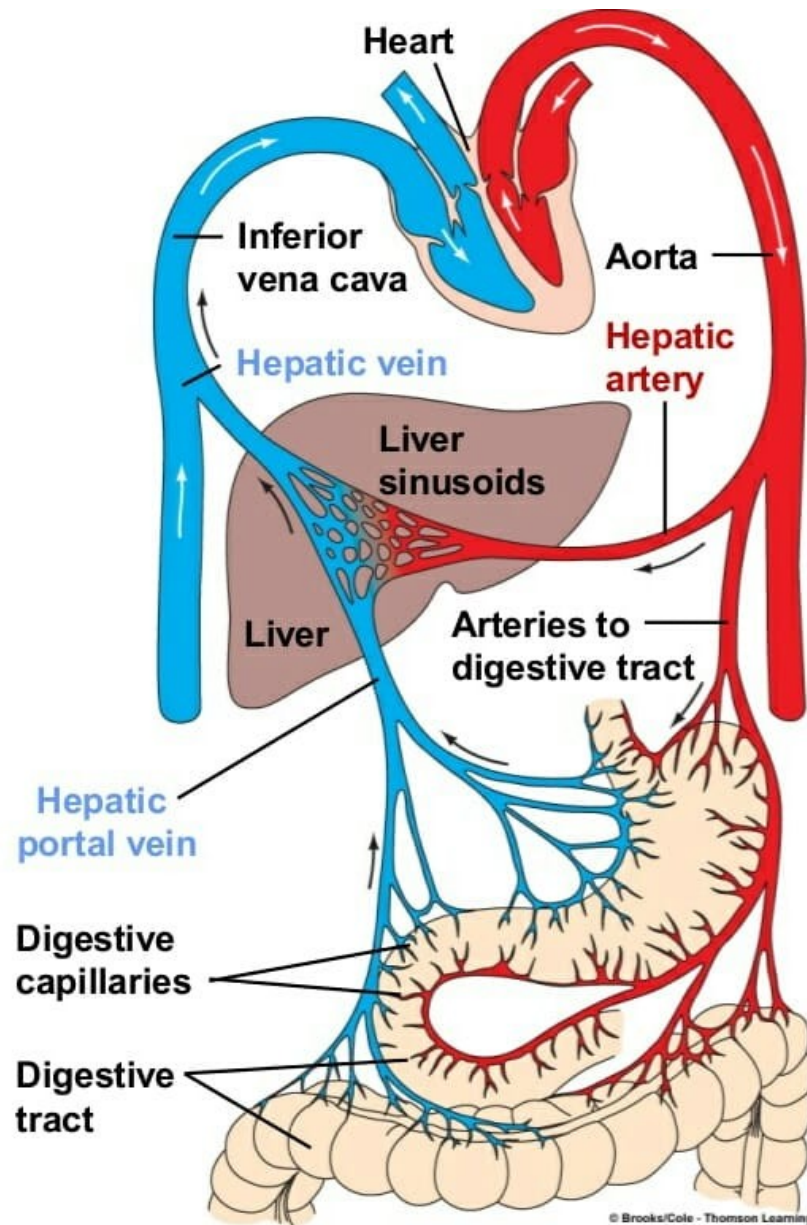
- parallels the abdominal aorta on the right as it ascends through the abdominal cavity.
- It penetrates the diaphragm at T8 and empties into the **right atrium**
- in the abdomen it has tributaries corresponding to the branches of the abdominal aorta such as:
 - paired lumbar veins
 - paired renal veins
 - right and left gonadal veins
 - right and left suprarenal veins
 - inferior phrenic vein
 - Hepatic veins
 - common iliac veins



Hepatic portal vein

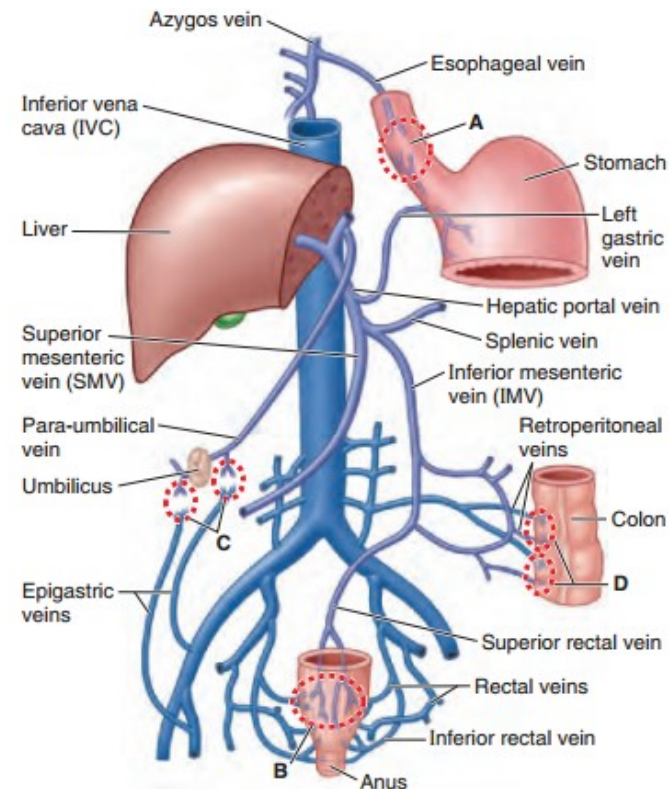
- It is formed behind the neck of the pancreas by union of **splenic** and **superior mesenteric veins**.
- Hepatic portal vein is a blood vessel that sends nutrient rich blood from the **gastrointestinal tract** and **spleen** to the liver, but also delivers toxins to the liver that will be chemically modified and filtered in the process of **detoxication**.

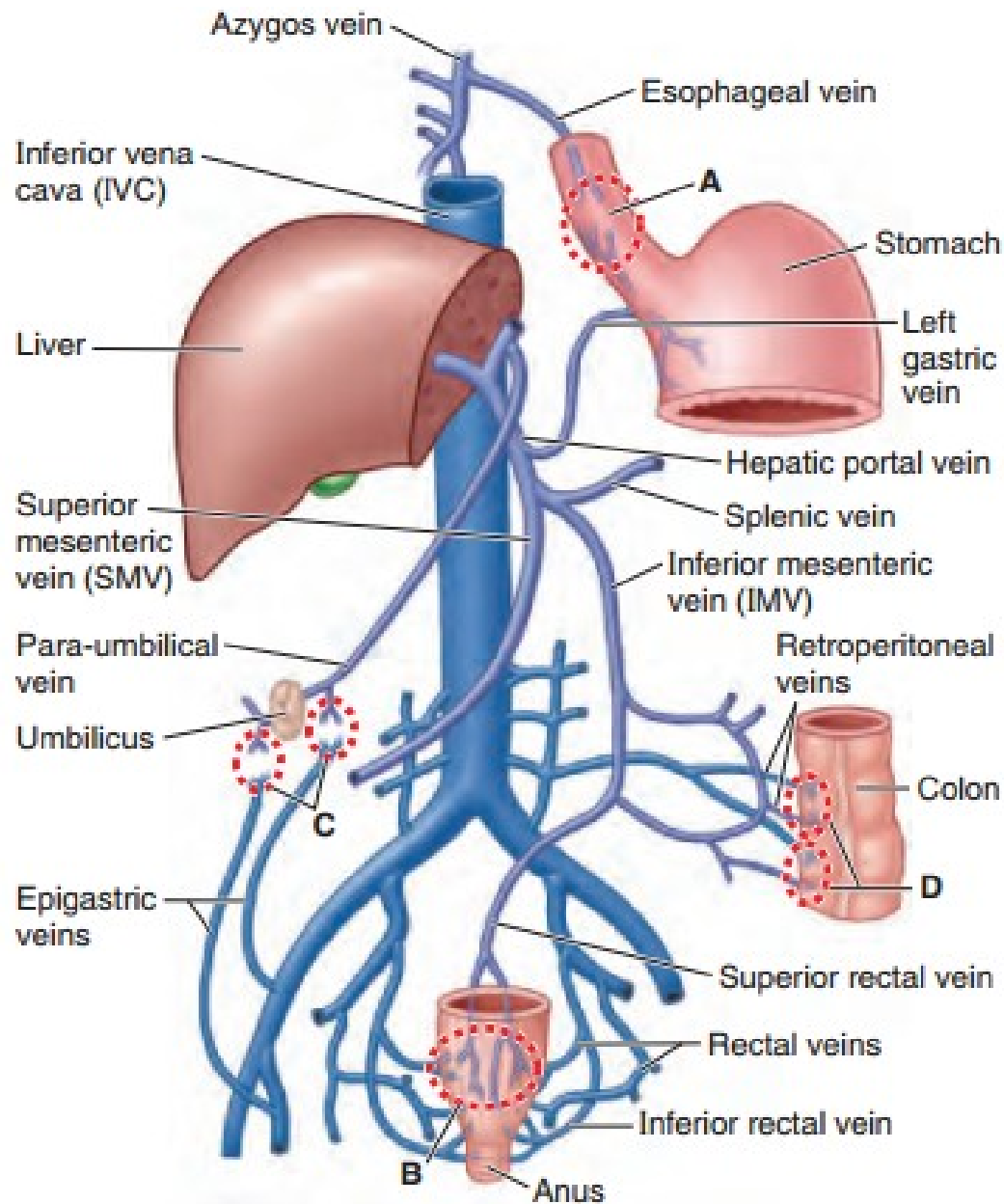




Portal–systemic anastomoses

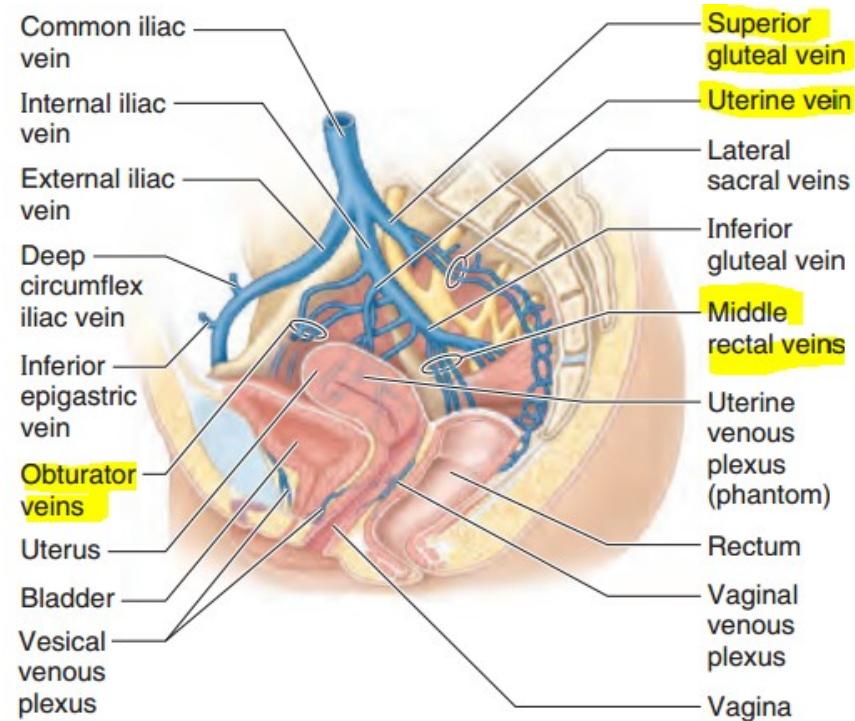
- Is in which the portal venous system communicates with the systemic venous system, are formed
 - in the inferior esophagus,
 - in the anal canal,
 - in the paraumbilical region, and in the liver.
- When portal circulation through the liver is diminished or obstructed because of liver disease or physical pressure from a tumor, blood from the this systems can still reach the right atrium through the **IVC** by way of these collateral routes.
- However, the **volume** of blood forced through the collateral routes may be excessive, resulting in potentially fatal *varices* (abnormally dilated veins)





Pelvic vein

- Tributaries of the internal iliac veins are more variable than the branches of the internal iliac artery with which they share names, but roughly accompany them, draining the same territories that the arteries supply.
- The **superior gluteal veins** are the largest tributaries of the internal iliac veins.



THANK YOU